

**A Web-Based Centralized Real-Time Monitoring and Integrated Booking Support
System for Fairfly Franchises**

A Capstone Project Proposal Presented to the Faculty of the
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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

Introduction

The travel and tourism industry experienced rapid growth and digital transformation due to advancements in information technology in recent years. The way reservations are handled has been completely transformed by technology. Booking flights, hotels, tours, and rental cars has become more automated and efficient thanks to Computer Reservation Systems (CRS) and Global Distribution Systems (GDS). This enhances overall efficiency by reducing errors and accelerating transactions (Hussain, 2024). However, these enterprise-level solutions are often developed for airlines, hotel chains, and large travel businesses, leaving small to mid-sized travel agencies without easily accessible and cost-effective tools tailored to their specific operations and scale.

This gap is well documented in the tourism digitalization literature. A systematic review of digital transformation research in tourism found that adoption barriers such as inadequate infrastructure, digital skill gaps, and financial constraints unreasonably affect small and medium-sized tourism enterprises compared to larger, resource-rich firms (Gutierriz, Ferreira, & Fernandes, 2023). This is consistent with more recent empirical evidence: in a study of tourism micro-firms and SMEs across 27 countries, using survey data from the European Commission, firms that lacked the skilled workforce needed for growth or faced financial constraints showed significantly lower odds of performance growth, while firms that both adopted digital technologies and defined a clear digitalization strategy saw the strongest gains (Succurro, 2026). Notably, the same study found that having a deliberate digital strategy mattered more for growth than simply adopting digital

tools in isolation suggesting that ad hoc digitalization, without a coherent plan, delivers weaker results than a structured approach.

Despite the industry's growing digitalization, many travel companies still fail to adopt digital solutions. According to Patricius IT (2025), over 57% of travel agencies are still managing expenses manually, which leads to inefficiencies, greater operational costs, and frequent errors. Employees in these agencies commit an estimated 40–60% of their time to repetitive administrative tasks rather than revenue-generating activities, and 5–10% of manual processing errors result in commission disparities and service-related concerns. Over time, these inefficiencies compound: staff hours that could otherwise be spent on client acquisition or service quality are instead consumed by paperwork, and recurring processing errors gradually erode both profitability and client trust.

These challenges become considerably more complex once an agency begins to franchise. Franchising allows a business to expand its market reach without the capital burden of opening and staffing new branches directly, but it introduces new coordination demands: the franchisor must maintain consistent service quality, pricing, and reporting standards across every branch, while franchise operators need clear, standardized procedures to follow from day one. Without a centralized system, franchisors struggle to track the sales, expenses, and service quality across multiple locations, often relying on decentralized spreadsheets and manual reporting that delay critical insights and make it difficult to identify underperforming branches before problems escalate (Amalraj, 2025). In practice, this means a franchisor may only discover a branch's declining sales or a recurring service issue weeks after the fact, by which point client relationships and brand reputation may already be affected.

Fairfly Travel and Tours agency faced similar issues. As the travel agency sought growth through franchising, its reliance on office-based systems and dispersed record-keeping became a barrier to its progress. Franchise operators lacked a uniform system for processing bookings and following operational protocols, and the head office had limited access to each branch's day-to-day activity. As Fairfly onboarded new franchise branches, the very inconsistencies described above such as decentralized reporting, delayed insight into performance, and the absence of a standardized workflow began to surface within its own operations, threatening the brand consistency and service quality the agency had built over years of face-to-face service.

To address these challenges, this study developed the design and development of a Web-Based Centralized Real-Time Performance Monitoring and Integrated Booking Support System for Fairfly Franchises. This study integrated booking management and franchise performance monitoring into a single platform. Unlike existing travel systems that primarily catered to customer-facing bookings, this system focused on operator-driven transactions and franchise-level management, addressing a gap that was largely unmet by current industry solutions.

Project Context

Fairfly Travel and Tours is a travel agency located in Baliuag, Bulacan. Founded in 2019, the agency is supported by its owner's more than two decades of expertise in the travel and tourism sector, providing a solid and strong foundation of industry knowledge and credibility. At present, Fairfly operates from a single physical branch, relying heavily on manual and office-based transaction processes. The agency does not currently use a web-based system to manage bookings, inquiries, or customer records. While this strategy has kept everyday operations running, it has considerable limitations in terms of efficiency, accessibility, and client reach.

Fairfly's long-term goal is to grow its client base and improve its service delivery by implementing digital solutions that allow clients to access services online. The agency needs a system that enables clients to inquire, select services, and submit their details remotely, while giving operators the tools to assess and process these requests efficiently.

At present, Fairfly offers several travel and documentation services, which defined the operational scope that the developed system supported. The agency provides Visa Assistance, which involves guidance and support in completing visa applications, including document preparation and submission for various destinations. It also offers Tour Packages, consisting of curated domestic and international tour packages that include transportation, accommodations, and itinerary planning. Additionally, the agency handles Airport Travel Booking, assisting clients in booking domestic and international flights with convenient scheduling and reliable travel arrangements. Fairfly also manages Hotel and Resort Reservations by coordinating with partner accommodations to secure comfortable and affordable lodging options. Furthermore, the agency provides Passport Processing

Assistance, helping clients prepare and submit passport application requirements to ensure accurate and timely processing. Lastly, the agency arranges field trips and team-building activities by managing transportation, accommodations, and travel coordination for educational and corporate group events.

All of these services are currently processed manually through face-to-face transactions at the physical branch. There is no digital system for clients to submit inquiries, choose services, or input their details online. As Fairfly seeks to improve its service delivery and expand its client reach, this manual operation creates a critical gap where clients cannot access the agency's services remotely, and the operator has no standardized workflow for processing client requests.

Purpose and Description

This capstone project designed and developed a Web-Based Centralized Real-Time Monitoring and Integrated Booking Support System for Fairfly Travel and Tours Franchises. The system served as the digital backbone of Fairfly's franchise operations, transforming its current manual, office-based processes into an accessible online platform that seamlessly connects franchise operators to the head office under a centralized and standardized environment.

The primary objective of this project is to convert Fairfly's existing internal operations into a unified online platform that primarily caters to business partners and franchise operators rather than individual travelers. These franchisees were able to manage their own branches under the Fairfly brand while remaining fully connected to the main office. Through this centralized system, franchise operators had standardized access to procedures and operational guidelines. They were able to manage bookings for local and

international travel services, including flight ticketing, visa assistance, tour packages, and other related arrangements. The system also enabled the automatic generation of sales and booking reports, eliminating the need for manual preparation, while ensuring that all records and transactions are securely stored within the unified Fairfly System.

At the same time, the head office or main branch of Fairfly Travel and Tours have full administrative access to the platform. This allowed them to monitor the performance and sales of each franchise branch in real time, review reports, transactions, and booking activities, maintain control over service standards, pricing structures, and operational consistency, and provide updates and support to franchise branches through a centralized communications channel.

Research highlights that digital systems play a critical role in enabling scalability and coordination in tourism enterprises, particularly when businesses expand to multiple locations (Vlahović et al., 2024). Centralized online platforms have been shown to improve operational efficiency, enhance communication, and ensure quality control in distributed business models such as franchising (World Tourism Organization [UNWTO], 2023). Furthermore, digital transformation significantly improves transparency, reporting accuracy, and management oversight in tourism businesses (Diaz-Castillo et al., 2025).

This project is significant because it goes beyond developing a simple travel booking website. Instead, it creates a structured online management environment that directly supports Fairfly's franchising goals. By transitioning from an office-based system to a widespread online platform for its franchisees, Fairfly will be able to expand to new locations without the need for constant physical supervision, standardize operations across branches, enable real-time monitoring and performance evaluation, reduce dependence on

manual reporting and face-to-face training, and strengthen its competitiveness in a digitally driven tourism industry. Studies confirm that tourism businesses that successfully integrate digital systems into their operations are better positioned for sustainable growth (UNWTO, 2023; Vlahović et al., 2024). As the global tourism industry increasingly relies on digital coordination and online platforms, a centralized, franchise-ready system is no longer just an advantage, it has become a necessity.

Project Objectives

General Objective

To design and develop a Web-Based Centralized Real-Time Performance Monitoring and Integrated Booking Support System for Fairfly Travel and Tours that standardizes franchise operations through template-driven booking workflows, centralizes administrative oversight, and enables real-time performance monitoring to support franchise expansion.

Specific Objectives

1. To analyze the current manual processes and operational workflow of Fairfly Travel and Tours to identify areas for digital improvement.
2. To develop a Head Office (Administrator) Module with the following features:
 - 2.1. An authentication form for administrators.
 - 2.2. A franchise management interface for registering, activating, and suspending franchise branch accounts.
 - 2.3. A booking template management system that allows the administrator to create, edit, and delete standardized booking forms and checklists,

including the designation of third-party provider links for each service type.

- 2.4. A centralized dashboard for real-time monitoring of Key Performance Indicators (KPIs) such as total sales and booking volume across all franchise branches.
 - 2.5. An inventory and content management feature for creating, updating, and deploying standardized tour packages and pricing to all branches.
 - 2.6. An integrated communication and support module for receiving technical or operational tickets to the head office.
 - 2.7. An event notification to notify and schedule an appointment when someone wants to apply for franchising.
3. To develop a Franchise Operator Module with the following features:
 - 3.1. A template-driven booking workflow interface that guides operators through step-by-step service processing, including redirection to designated third-party booking platforms and transaction detail recording.
 - 3.2. An integrated communication and support module for raising technical or operational tickets to the head office.
 - 3.3. A digital inquiry form to record inquiries from potential clients.
 - 3.4. A digital quotation form to record and notify the client of their requirements needed, tour dates, inclusions, exclusions, rates, and total amount of their availed service.
 - 3.5. An authentication form for franchise operators.
 4. To develop a Customer Module with the following features:

- 4.1. An AI-powered inquiry chatbot that draws responses from the system's database to assist potential clients with service information, requirements, and branch location inquiries.
- 4.2. A digital appointment form for client to schedule appointment for face-to-face consultation and inquiry.
- 4.3. A real-time tracker to track their availed service if it is still pending or completed.
- 4.4. An authentication form for customers.
- 5. To develop a franchise form for users to schedule appointment and apply for Fairfly franchising.
- 6. To evaluate the system's performance, usability, and reliability through ISO/IEC 25010.

Scope and Limitations

This study focused on the design and development of a Web-Based Centralized Real-Time Monitoring and Integrated Booking Support System for Fairfly Travel and Tours Franchises. The system aims to migrate the agency's current manual, office-based operations into a centralized online environment that supports business scalability, effective franchise monitoring, and standardized service delivery across all branches.

Scope of the Study

The system consists of three main modules: the Head Office (Administrator) Module, the Franchise Operator Module, and the Customer Module.

The Head Office (Administrator) Module includes franchise management functions such as the ability to register, activate, or suspend franchise branch accounts. It features a centralized dashboard that provides real-time data visualization of key performance indicators (KPIs), including total sales and booking volumes across all franchise branches. Administrators can also manage booking templates by creating, editing, and deleting standardized templates for each service type. These templates define the required customer information fields, the step-by-step processes that operators must follow, and the designated third-party booking platforms to be used at each stage. Additionally, the module offers an inventory and content management system (CMS) for creating, updating, and deploying standardized tour packages and pricing across all branches. A resource library management tool allows the uploading and organization of operational guidelines, training videos, and marketing assets such as posters and banners. Finally, a technical support ticketing system enables the administrator to monitor and respond to support tickets raised by franchise branches regarding operational or technical issues.

The Franchise Operator Module provides a template-driven booking workflow that guides operators through each step of processing customer transactions. This interface follows the standardized booking templates created by the administrator, directing operators to fill in customer information, redirecting them to designated third-party platforms for external reservations, and allowing them to record transaction details before proceeding. The module also includes integrated communication and support features,

enabling operators to raise support tickets directly to the head office for technical or operational assistance. Furthermore, operators have access to a digital resources library where they can retrieve the latest training materials and marketing resources distributed by the head office, eliminating the need for external file-sharing tools.

The Customer Module features a public-facing inquiry interface with an AI-powered chatbot. This landing page allows potential clients to interact with the chatbot, which draws responses from the system's database to provide information about available services, requirements, and branch locations. The chatbot then directs inquiries to the nearest franchise branch for further operator-assisted service.

The system was developed as a web-based platform accessible through standard web browsers to ensure cross-platform compatibility. It was built using React.js for a fast, responsive, and dynamic user interface, and will utilize Firebase for real-time and secure database management.

Limitations of the Study

The following constraints define the functional boundaries of the system, considering the development timeline and technical scope:

The system is designed as a Business-to-Business (B2B) and Business-to-Client (B2C) platform. While it includes a public inquiry landing page with an AI chatbot, all actual booking transactions are restricted to the operator-side portal. Customers cannot directly book or purchase services through the system. In addition, the system does not feature direct API integration with third-party booking platforms. Instead, operators are redirected to external websites designated by the administrator to complete reservations manually, after which they record the transaction details in the system. Live payment

processing is also not supported; the system can only record and track payment statuses (such as “Paid,” “Pending,” or “Partial”) for reporting purposes, but will not handle actual electronic payments or integrate with gateways like Maya, GCash, or PayPal.

The AI-powered chatbot is limited to handling general inquiries about services, requirements, and branch locations based on data stored in the system’s database. It is not designed to process transactions, provide personalized travel recommendations, or replace human agents for complex customer queries. As a cloud-based web application, the system requires a stable internet connection to function and does not support offline mode or local database synchronization. Any interruption in connectivity could prevent franchise operators from accessing real-time data or processing new bookings. Finally, the study is geographically limited to the operations of Fairfly Travel and Tours and its franchise branches within the Philippines.

Definition of Terms

AI Chatbot - An artificial intelligence-powered conversational interface embedded in the system's customer-facing landing page that provides automated responses to inquiries based on data stored in the system's database.

B2B (Business-to-Business) - A commercial model involving transactions between businesses. In this study, it refers to the operational relationship between the Fairfly Head Office and its franchise operators, as opposed to direct customer-facing transactions.

Booking Template - A structured form and step-by-step checklist created by the administrator for a specific service type. It defines the required customer information

fields, the sequential process the operator must follow, and the designated third-party booking platforms to be used at each stage.

Centralized Platform - A unified system architecture in which all franchise branch data, transactions, reports, and user accounts are stored and managed within a single main database accessible by the Fairfly Head Office.

CMS (Content Management System) - A system feature that enables the administrator to create, update, and deploy standardized content such as tour packages, pricing, and service offerings across all franchise branches.

Dashboard - A graphical interface within the system that displays real-time data and Key Performance Indicators (KPIs), allowing the administrator to monitor franchise performance, sales, and booking activity.

Firebase - A cloud-based platform developed by Google that provides real-time database management, authentication, and hosting services. It serves as the backend infrastructure for the developed system.

Franchise - A business arrangement in which an independent operator (franchisee) is authorized to conduct business under the Fairfly brand, following standardized operational procedures and guidelines set by the head office (franchisor).

Franchise Operator - An individual or entity that manages a Fairfly branch location, utilizing the system's template-driven booking workflow and tools to process customer transactions under the supervision of the head office.

KPI (Key Performance Indicator) - A measurable value used to evaluate the performance of a franchise branch. In this system, KPIs include metrics such as total sales, booking volume, and transaction completion rates.

React.js - An open-source JavaScript library developed by Meta (formerly Facebook) used for building fast, responsive, and dynamic user interfaces. It serves as the front-end framework for the proposed system.

Resource Library - A digital repository within the system where the head office uploads and organizes operational guidelines, training materials, and marketing assets for access by franchise operators.

Scalability - The capacity of the system to accommodate additional users, franchise branches, and transaction volumes without compromising functionality or performance.

Template-Driven Booking Workflow - The system's core operational process in which franchise operators follow administrator-created booking templates to process customer transactions step by step, including redirection to designated third-party booking platforms for external reservations.

Third-Party Booking Platform - An external website or service (e.g., airline booking sites, hotel reservation platforms) to which operators are redirected during the booking workflow to complete specific reservations. The system does not integrate with these platforms through APIs; operators complete the external booking manually and record the transaction details upon return.

Web-Based System - An application accessible through standard internet browsers that enables centralized management of transactions, data storage, and operational processes without requiring local software installation.

CHAPTER II

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter presents a review of related literature and systems that are relevant to the developed system. The purpose of this chapter is to examine existing research, concepts, systems, and technologies that support the design and implementation of a web-based centralized real-time monitoring and integrated booking support system for Fairfly Travel and Tours Franchises.

Related Literature

Digital Transformation in Tourism and Travel Agencies

According to Margatan (2025), the move to digital transformation allows travel agencies to increase their business efficiency and reduce daily operational costs. This research suggests that utilizing tools like e-commerce and big data is not only for marketing but also for creating more professional opportunities for small agencies to compete effectively. For a franchise model, a B2B system provides "participation depth," which means branch operators can utilize the head office's resources to perform more effectively than if they worked independently (Miao & Yu, 2026).

Research on travel agency readiness also indicates that many agencies are increasingly shifting from manual operations to integrated online booking platforms as part of their digital transformation. This transition reflects a growing recognition that automation is necessary to improve operational efficiency, reduce errors, and enhance competitiveness in the travel industry. The study emphasizes that readiness to adopt such

systems demonstrates the industry's awareness of the benefits of replacing traditional, manual processes with unified and automated digital solutions (Aziz et al., 2023).

In addition, Milojković et al. (2024) found that travel agencies that use digital tools become more competitive and efficient and suggest that travel agencies in developing nations undergo a digital transformation by offering online services via their websites or mobile applications; process automation through the use of software solutions for processing reservations, issuing vouchers and invoices, inventory management, and similar tasks. In relation to the present study, developing a web-based, centralized, real-time monitoring and booking support system for Fairfly can help the agency improve scalability and become more competitive in the industry.

Customer Satisfaction in Physical vs Online Transactions

Customer satisfaction is an important measure of service quality, influenced by how well a business meets its clients' expectations. When it comes to travel agency services, physical processes often lead to higher satisfaction because customers receive personalized and immediate guidance, and most especially for providing reassurance on matters involving financial transactions. Accordingly, the following review of related literature supports this topic.

According to Nguyen (2024), although online travel platforms offer convenience, customers often prefer physical interaction when handling complex travel arrangements. Travel services involve multiple components, such as itinerary planning and visa processing, which are easier to manage through direct communication. Face-to-face interaction enables agencies to provide immediate assistance and reassurance, leading to increased customer satisfaction and trust.

In line with this, Fairfly recognizes the advantages of physical processes based on its experience in handling travel services. Its hands-on approach enables real-time communication, ensures accuracy, and enhances customer confidence. This supports the continued use of in-person services for improved reliability and a more personalized customer experience (Nguyen, 2024).

Franchising Business and Technology Adoption

Several studies highlight the increasing adoption of digital platforms in travel agencies and tourism enterprises. Abdul Ghani et al. (2022) explain that franchising enables businesses to expand into new markets by leveraging an established brand and proven business model, ensuring franchisees can start with lower risk than independent startups. The study identified key growth factors for franchising, including product and service innovation, strong franchisor–franchisee relationships, and government support, which collectively contribute to business scalability and sustainable growth in service sectors. A study by Li et al. (2025) demonstrates that online booking systems reduce operational costs, improve customer satisfaction, and enhance coordination across departments. These findings support the proposed implementation of web-based travel management systems that streamline booking operations and coordination among employees and franchise branches.

Sabaran and Abd Aziz's (2023) study emphasizes the importance of technology to day-to-day franchise operations, supporting branch coordination, management, and communication. Franchises use technology to boost productivity, improve data quality, expedite communication, and gain a competitive edge in areas where digital solutions are to be expected.

In the study by Galván-Sánchez et al. (2021), the findings indicate that, in general, franchise systems use websites to manage their relationships with stakeholders, such as customers and franchisees. To help with digital communication, several franchises have a press room on their websites where they post corporate news about the franchise, which is helpful when talking to other stakeholders. In relation to the development of a web-based centralized system, having a franchising system will enhance the operations and communication of Fairfly agency.

Web Development Technologies

In Web-Based Business Systems, Performance is a vital element, especially for those that support multiple branches and that require real-time monitoring, such as Franchising and Management Systems. If a system loads slowly or has frequent major delays in data updates, it can negatively affect the productivity of the business, reporting activity, and user satisfaction.

Recent studies comparing modern web development frameworks have shown that React.js is efficient in managing dynamic UI (User Interface) that can handle multiple and frequent data updates (Piastrou, 2023). Because the Fairfly system includes dashboards, management, and monitoring tools, efficient rendering of information in the UI is essential. Recent research has also emphasized that proper and clean component structuring and organized coding practices can significantly improve system responsiveness and reduce unnecessary processing (Veeri, 2024).

Cloud Computing and Security

Business-to-business (B2B) and franchising systems can handle sensitive and critical operational, financial, and performance data and information across multiple branches of the business, as for the proposed system, franchise operators and the head office will access shared booking records, operational documents, and sales reports. Securing these data is essential to maintain the CIA Triad (Confidentiality, Integrity, and Availability) and Trust among the business partners.

Studies on systems that are using cloud technologies have emphasized that even though service providers provide the infrastructure needed, the organizations are responsible for managing the data access and user permissions properly and cleanly (Demissie, 2021). In B2B businesses, having clearly defined roles for the users of the systems, such as franchise operators and administrators, is necessary to ensure that each user of the system can only access authorized information.

A systematic review on cloud computing security has shown and identified unauthorized access and system misconfiguration as the major causes of data breaches and system mishaps (Alouffi et al., 2021). In systems such as the proposed franchising system for Fairfly, multiple branches will connect to a centralized database, and improper configuration may expose critical sales or financial records of the business. This study highlights the importance of robust authentication processes and well-defined access controls.

Artificial Intelligence in Hospitality and Customer Experience

The use of Artificial Intelligence (AI) in the Philippine tourism industry has grown rapidly, especially for handling customer inquiries. Recent research in the Philippines highlights that Filipino travelers increasingly prefer digital solutions for their initial inquiries. A study by Espiloy (2025) found that chatbot responsiveness and AI-driven personalization are one of the strongest predictors of customer satisfaction and loyalty in the hospitality sector. This research shows that when a system can give instant, accurate information, it builds trust with the customer before they even make a booking. Furthermore, Miao and Yu (2026) argue that these digital tools are essential for small agencies because they provide "participation depth." This allows even small franchise branches to handle a high volume of inquiries through the head office's AI infrastructure, which helps them compete with much larger travel brands by co-creating value through the platform.

In a Business-to-Business (B2B) context, the AI chatbot serves as a digital filter between the customer and the business partner. Academic studies on "AI-driven CRM systems" (Sirivadhanawaravachara, 2025) explain that AI can learn about customer habits and preferences during the inquiry stage by analyzing their interactions. This data is then passed to the human operator, which helps the franchisee provide a more tailored service when they finally talk to the client. This "hybrid" approach, where AI handles the initial talk and humans handle the final booking, is seen as the most reliable way to protect business margins while keeping customers happy.

Template-Driven Workflow Systems and Process Standardization

The proposed system's core operational mechanism is a template-driven booking workflow, where franchise operators follow structured, step-by-step processes created and managed by the head office. This approach is grounded in the established concepts of workflow automation and process standardization.

Workflow automation enables organizations to streamline business processes by defining structured sequences of tasks that execute according to predefined rules. Mohan et al. (2022) define workflow automation as the process of making tasks, documents, and information flow independently in accordance with defined business rules, reducing manual intervention and ensuring procedural compliance. In the hospitality sector, Goyal and Singh (2021) demonstrated that implementing workflow automation in customer service operations reduces human errors and improves overall efficiency by guiding staff through structured, step-by-step process flows. These findings support the design of the proposed system, where admin-created booking templates guide franchise operators through each stage of a customer transaction.

In the context of franchising, process standardization through digital tools is critical for maintaining operational consistency across distributed branches. Nazarkevych (2025) examined how modern digital technologies enable process automation and standardized operations within franchise networks, emphasizing the role of integrated software in ensuring control and interaction between franchisors and franchisees. Similarly, Peskova (2025) highlights that digital tools simplify and optimize franchise operations by enhancing communication, collaboration, and procedural uniformity between the franchisor and its branch operators. Sabaran and Abd Aziz (2023) further emphasize that technology is

essential to day-to-day franchise operations, supporting branch coordination, management, and communication while boosting productivity and data quality.

The importance of standardized procedures in digital business environments is further supported by Meisarandi and Kurniawati (2024), who found that implementing Standard Operating Procedures (SOPs) in a digital team environment ensures operational consistency, compliance, and accountability across all members of the organization.

These studies collectively provide the theoretical foundation for the proposed system's template-driven approach. By enabling the head office to create, modify, and deploy structured booking templates that guide franchise operators through each transaction step, the system applies workflow automation and SOP principles to ensure operational consistency, reduce processing errors, and maintain service standards across all Fairfly franchise branches.

Scrum Framework in Software Development

Agile methodology with the Scrum framework has become increasingly popular in the development of software systems for the tourism and travel industry due to its iterative nature, flexibility in handling changing requirements, and ability to deliver functional increments rapidly. Scrum emphasizes collaboration, transparency, and continuous improvement through defined roles (Product Owner, Scrum Master, and Development Team), sprints, daily stand-ups, sprint reviews, and retrospectives. This approach is especially suitable for projects requiring centralized control, standardized processes, and real-time monitoring features that align closely with the needs of a franchised travel agency system.

Anifa et al. (2024) conducted a systematic review of literature on the agile approach and found that agile methods, including Scrum, are particularly effective in addressing complexities in modern business environments. The review emphasized Scrum's advantages in promoting scalability, better communication, transparency, and adaptability—qualities essential for tourism businesses transitioning to franchise models with centralized oversight and standardized operations.

El Hakim et al. (2023) implemented the Scrum framework in building an application for monitoring and booking e-buses based on QR codes. The project utilized standard Scrum roles and practices to develop a prototype that combined monitoring dashboards with booking capabilities. Their experience underscores Scrum's strength in creating systems with centralized tracking and operator-guided workflows, which mirror the Head Office dashboard and Franchise Operator Module in the proposed Fairfly system.

Mixed-method in Data Gathering

Mason et al. (2021) stated that the systematic use of mixed methods in research, arguing that an initial qualitative phase can effectively inform a subsequent quantitative phase. This sequential approach allows researchers to explore user needs and operational challenges qualitatively before validating them through surveys and such, making it highly suitable for capstone projects involving the design and evaluation of centralized booking and monitoring systems for franchise operations.

The adoption of mixed-methods research is justified for this project because it allows for a strong evaluation of the proposed Web-Based Centralized Real-Time Monitoring and Integrated Booking Support System. Qualitative data from Fairfly management can uncover contextual challenges, user satisfaction, and areas for

improvement while quantitative data from Fairfly employees and IT experts can measure operational efficiency and franchise performance. This combination strengthens the study's credibility, addresses both technical and human factors, and provides actionable recommendations for Fairfly Travel and Tours' proposed system.

Related System

Franchise Management Systems

The study of Shejwal et al. (2025) states that a franchise management system is a modern way to solve problems that franchise companies, like travel booking businesses, have with their day-to-day operations. The system lets users make smart choices, improves transparency, and encourages better teamwork by automating tasks, making communication safe, and giving users access to real-time data. Smart automation and modern online technology can change how franchises work around the world with a central franchise management system. The centralized system helps the franchising business save money, get more done, make sure that all locations follow the same rules, and improve service standards. As digital transformation happens, this unified platform could change the world of franchising.

A software development case study by Flairstech (2025) has shown that they transformed a restaurant chain's existing manual processes into a CRM (Customer Relationship Management) based and powered franchise management system for real-time sales tracking and compliance for other branches spread across locations, which, in the study, helped the chain achieve 40% efficiency gains. This case study can be applied directly to Fairfly, extending its internal main branch system to a franchise-wide centralized monitoring and management for future scalability and expansion.

E-Booking and Reservation Systems

With the rise of digital transformation in the tourism industry, many agencies now rely on online systems for their business operations. From the perspective of the customer, the idea of web-based systems such as e-booking is straightforward and practical, in addition to saving time and money. Digital services are becoming increasingly important to both businesses and customers, making them one of the best features of modern tourism (Vlahović et al., 2024). These findings highlight the importance of web-based systems in travel agencies and reinforce that developing an integrated booking system for Fairfly is necessary to support centralized monitoring and management for future scalability and expansion.

Web-Based Systems Using React and Firebase

In the last five years, many enterprise web systems have shifted toward cloud and serverless setups to better handle scaling, security, and maintenance. According to recent studies, React.js and Firebase are a well-liked tech stack for creating quick, real-time apps. Systems with multiple user roles, such as franchise platforms, benefit greatly from React's component-based structure, which makes it simple to create modular user interfaces. In addition, Firebase offers a backend-as-a-service that handles authentication and databases without needing a traditional server.

According to a study by Yerlekar et al. (2023), React.js and Firebase may be used together to create complete projects, such as an e-commerce design, where Firebase services offer backend assistance like database access and cloud hosting while React handles front-end interactivity. Applications developed in this manner have interactive

behavior that enables asynchronous data uploading and downloading, which keeps user interface elements responsive and in sync with backend data without needing page reloads

A study by Kumar et al. (2025) on the CTMS (Comprehensive Travel Management System) uses Google's BaaS (Backend as a Service) Firebase, enabling agencies that do travel booking and management to coordinate client group trips, flights, and hotels with matching that is backed up by algorithms and multi-user access, which in the study says reduces manual errors in fragmented operations to those that have no payment processing. This study supports the centralized franchise and its monitoring, like the proposed Fairfly's B2B or Franchising model, by standardizing data across its upcoming branches for more scalable insights.

Another study by Sagar (2024) utilized React for front-end development and Firebase Authentication and Firestore for backend services to create a real-time chat application. This approach highlighted how Firebase's backend capabilities support dynamic data flows in web applications and demonstrated that utilizing cloud-based real-time databases to enable synchronous and collaborative user interaction without manual refreshes is both feasible and scalable. In addition to front-end performance, as is the case with React.js, database design significantly affects and determines system speed and overall reliability. Research on Firebase optimization has shown that well-organized and structured data, along with controlled and stable data retrieval, has reduced response delays and improved overall system efficiency (Semma et al., 2022).

Digital Monitoring and Dashboard Systems

Monitoring financial compliance in real time has emerged as a critical component of modern business operations. Continuous oversight of financial transactions is facilitated by real-time monitoring, enabling the identification of compliance violations promptly and the implementation of corrective action (Adekunle et al., 2024). These findings support the idea that a web-based, centralized, real-time monitoring system for Fairfly can improve business operations, efficiency, and transparency among franchise branches, thereby lowering errors and facilitating proactive management within the organization.

A 2025 case study by Urmeneta et al. demonstrated that the centralized management information system has effectively reduced data redundancy through the use of secure role-based login and an administrative side of the system that can serve as the main gateway for data management. This study can be applied to the Fairfly system, which implements role-based login for employees to secure important data and transactions within the centralized system. Ordóñez-Martínez et al. (2024) also explain that geo-dashboards in tourism management integrate continuously updated data from multiple sources, enabling real-time visualization of tourism activities that support timely decision-making and effective operational control. Similarly, the proposed web-based system includes an administrative dashboard that allows the main branch to monitor franchise activities and booking transactions in real time, improving operational oversight and management efficiency.

Comparison Matrix

To clarify how the developed system differs from existing solutions, Table 1 compares its core features against representative franchise management, CRM, and travel booking systems discussed in this chapter.

Table 1

Comparison of the Proposed System with Related Systems

Feature	Generic Franchise Management Systems (e.g., Shejwal et al., 2025)	CRM-Based Franchise Systems (e.g., Flairstech, 2025)	Travel/E-Booking Systems (e.g., Kumar et al., 2025)	Proposed Fairfly System
Real-time centralized dashboard	Yes	Yes	Limited	Yes
Built specifically for travel agencies	No	No	Yes	Yes
Template-driven booking workflow	No	Partial	No	Yes
Operator-driven (B2B) transaction model	No	Partial	No	Yes
Franchise-specific onboarding/application feature	No	No	No	Yes
AI chatbot for customer inquiries	No	No	Varies	Yes
Designed for small/mid-sized agencies	No	No	Varies	Yes
Direct third-party payment integration	Varies	Varies	Varies	No (by design/limitation)

As shown in Table 1, existing systems each address parts of Fairfly's operational needs, but none combine franchise-specific, template-driven booking workflows with real-time centralized monitoring in a system purpose-built for a small-to-mid-sized travel agency. This comparison further supports the rationale for the proposed system.

Synthesis

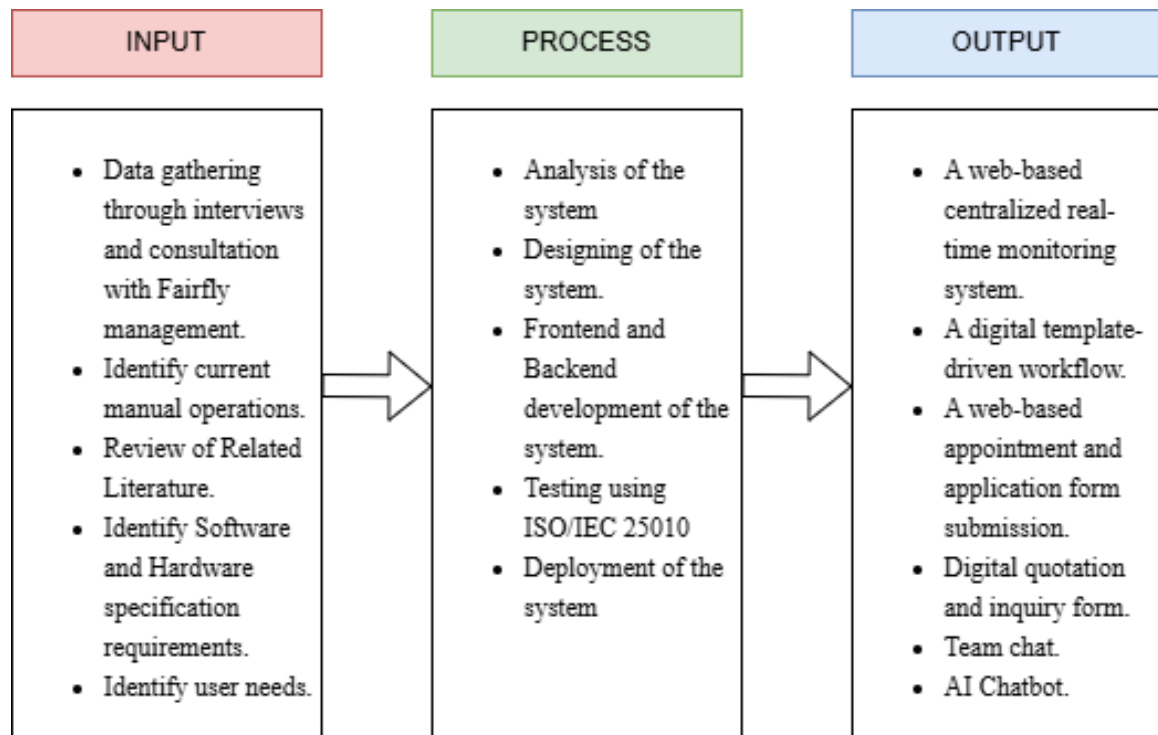
Taken together, the literature and related systems reviewed in this chapter establish three consistent findings relevant to this study. First, digital transformation measurably improves efficiency and competitiveness for travel agencies, but small and medium-sized agencies face structural barriers to adoption that larger enterprise-focused systems do not address (Margatan, 2025; Aziz et al., 2023; Milojković et al., 2024). Second, franchising research consistently shows that centralized digital coordination, standardized workflows, and clear franchisor-franchisee communication are what allow franchise networks to scale without sacrificing service consistency (Abdul Ghani et al., 2022; Sabaran & Abd Aziz, 2023; Shejwal et al., 2025). Third, the specific technologies proposed for this system, React.js and Firebase for real-time web applications, template-driven workflow automation for process standardization, and role-based access control for data security, are each independently supported by recent studies as effective solutions to the operational problems Fairfly faces.

However, the reviewed related systems reveal a consistent gap. Existing franchise management platforms and CRM-based systems (Shejwal et al., 2025; Flairstech, 2025) are largely built for large, established franchise networks and general retail or restaurant chains, not for small travel agencies operating a B2B, operator-driven booking model.

Similarly, existing e-booking and travel management systems (Kumar et al., 2025; Vlahović et al., 2024) are designed primarily for direct customer-facing transactions, not for supporting franchise operators who process bookings on a customer's behalf through third-party platforms. No single reviewed system combines a template-driven, franchise-specific booking workflow with real-time centralized monitoring built specifically for a small-to-mid-sized travel agency's franchise expansion. This is the specific gap the developed Web-Based Real-Time Monitoring and Integrated Booking Support System for Fairfly Travel and Tours Franchises addresses.

Conceptual Framework

This conceptual framework follows the Input-Process-Output (IPO) model, a functional graph that provides a structured overview of the research and development lifecycle. It illustrates how raw data is transformed through specific technical actions into a finalized web-based, centralized, real-time performance monitoring and integrated booking support system for Fairfly Travel and Tours. Figure 1 illustrates the IPO model of the developed system, mapping how raw data inputs are transformed through system development processes into the finalized web-based platform.

Figure 1*IPO model of the Developed System*

CHAPTER III

TECHNICAL BACKGROUND

This chapter presents the technical background and research methodology used in the development of the system. This chapter discusses the system requirements, research design, and the software development methodology adopted by the researchers. It also presents the system design through different modeling tools, such as diagrams and system representations that illustrate the flow of data, system interactions, and database relationships. Through these components, this chapter provides a comprehensive overview of the technical framework that guided the development and evaluation of the developed system.

Research Design

This study used a mixed-method research methodology that includes both qualitative and quantitative approaches to obtain and analyze data related to the developed system. An integrated mixed-methods approach can widen the evaluation's variety of perspectives on the technical development issues being studied. By combining the range and generalizability of quantitative approaches with the depth of understanding of processes, attitudes, and behavior given by qualitative approaches, it can also increase the precision of both the quantitative and qualitative components of the evaluation (Bamberger, 2022).

Qualitative Method

The qualitative approach was used throughout the information gathering stage, where interviews and consultations with the management of Fairfly Travel and Tours were

conducted. These interactions allowed the researchers to understand the current operational processes of the organization, identify the limitations and problems of their existing manual and office-based system, and determine the necessary features that were needed in the developed web-based platform.

Quantitative Method

The quantitative approach was applied during the system evaluation phase. A structured questionnaire was distributed to the management and employees of Fairfly to evaluate the performance of the web-based system based on specific criteria such as usability, reliability, efficiency, and functionality. A five-point Likert scale was used for all evaluation items.

The combination of qualitative and quantitative methods enabled the researchers to gain both in-depth insights and measurable evaluation results that ensured that the developed system met the operational needs of Fairfly Travel and Tours while also complying with standard software quality measures.

Software Development Methodology

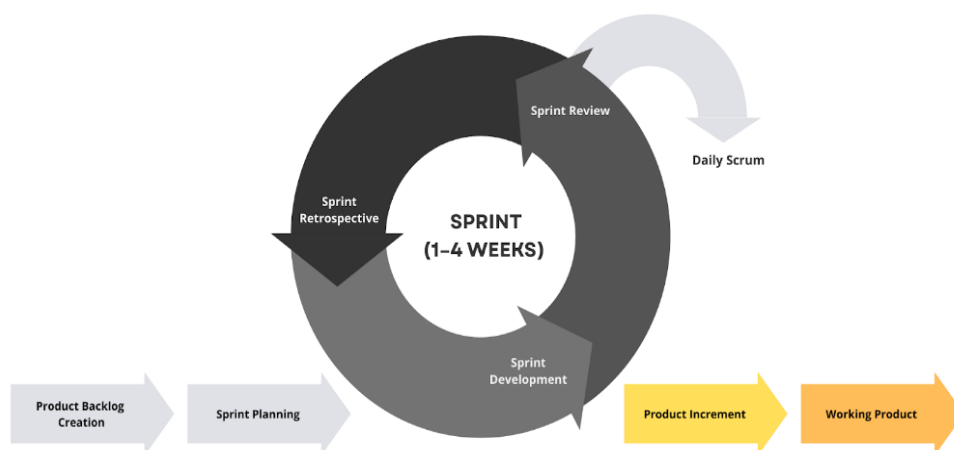
The researchers adopted the Agile methodology with the Scrum framework in the development of the proposed web-based system for Fairfly Travel and Tours. Scrum is a simple framework that enables individuals, groups, and companies to generate value by finding flexible solutions to challenging issues through iterative and incremental processes known as sprints, allowing teams to continuously inspect and adapt their work (Schwaber & Sutherland, 2020).

Agile Scrum breaks the development process into short phases called sprints, where certain system elements are created, developed, tested, and refined within a time period (Schwaber & Sutherland, 2020), in contrast to conventional linear and sequential development models like the Waterfall approach. This method enables developers to react swiftly to feedback and make the required changes as the project is being developed. The Agile Scrum framework was chosen for this developed system because it enabled the researchers to progressively develop system features while continuously assessing the platform's usability and functionality by routinely consulting the Fairfly management and employees. This framework guaranteed that the system changes in response to user demands and Fairfly Travel & Tours' operational requirements.

Figure 2 illustrates the Agile Scrum development process used for this study, outlining the iterative sprint cycle from sprint planning through sprint retrospective that guides the system's incremental development.

Figure 2

Scrum Process Diagram



As explained by Schwaber & Sutherland (2020) in the Scrum Guide, there are a total of three (3) main roles and three (3) artifacts in the Scrum framework. One Scrum Master, one Product Owner, and developers make up the Scrum Team. The Scrum team also employs Scrum artifacts, which are crucial pieces of information, to help define the product and outline the necessary steps to achieve it. A product backlog, a sprint backlog, and an increment with a definition of done (DoD) are the three Scrum artifacts (Atlassian, n.d.). There are also five (5) events or phases in a scrum sprint that are especially made to allow for the necessary transparency. Practices within the Scrum framework are employed to establish consistency and minimize the need for meetings that aren't specified in Scrum (Schwaber & Sutherland, 2020). be needed in the developed web-based platform.

Roles

- Scrum Master - Establishing Scrum according to the Scrum Guide (2020) is the responsibility of the Scrum Master. They accomplish this by assisting all members of the Scrum Team and the organization in comprehending Scrum theory and practice. The effectiveness of the Scrum Team is the responsibility of the Scrum Master. They accomplish this by allowing the Scrum Team to enhance its procedures within the Scrum framework.
- Product Owner - Product owners are concentrated on comprehending market, client, and business needs. This position sets the engineering team's work priorities. According to Atlassian (n.d.), successful product owners create and oversee the backlog of products, use sprint backlog management for efficient planning and prioritization by creating, organizing, and openly managing the product backlog, collaborates with the team and the company to make sure that everyone is aware of

the tasks in the product backlog, provides the team with precise instructions on what features to produce next, and determines when to ship the product, with a preference for more frequent delivery.

- Developers - They are members of the Scrum Team who are dedicated to producing any component of a usable increment during each sprint. They are responsible for the actual design, development, and testing of the system features during each sprint (Schwaber & Sutherland, 2020).

Artifacts

- Product Backlog - It is the major list of tasks that the product owner or product manager must complete and maintain. This is a dynamic list of features, requirements, enhancements, and fixes used to populate the sprint backlog (Schwaber & Sutherland, 2020).
- Sprint Backlog - It is a collection of items, user stories, or bug fixes picked up by the development team for implementation during the current sprint cycle. During sprint planning, the team selects things from the backlog to work on (Schwaber & Sutherland, 2020).
- The Increment - It is a concrete step toward the product goal. Each increment is thoroughly validated and adds to previous ones, assuring their seamless integration. To give value, the Increment must be usable (Schwaber & Sutherland, 2020).

Phases

Phase 1. Sprint planning is the first event that occurred in the sprint process. According to Atlassian (n.d.), during this meeting, the entire development team plans the scope of the work to be done during the current sprint. This meeting, called the sprint planning event, is led by the Scrum master. The team plans the work to be done and decides on the sprint goal during this meeting.

Phase 2. Sprint or Sprint Development is the next phase of the Scrum framework. The development team works on implementing the chosen features during the sprint. Usually lasting one to four weeks, a sprint is a time-boxed development cycle. The sprint's objective is to create a functional product increment that stakeholders can evaluate (Atlassian, n.d.).

Phase 3. Every day of the sprint, a brief meeting known as the Daily Scrum is held. Team members plan the work for the next 24 hours, discuss their progress, and identify challenges during this meeting. This keeps everyone focused on the sprint objective (Atlassian, n.d.).

Phase 4. Sprint Review is where the development team shows stakeholders the finished product at the conclusion of the sprint. Stakeholders can assess the product increment and offer suggestions for future development during the sprint review (Atlassian, n.d.).

Phase 5. Sprint retrospective is the last event before the product increment. This is where the team evaluates the development process, pinpoints its successes, and suggests areas that need improvement for the next sprints during this meeting. This encourages the team to keep improving (Atlassian, n.d.).

Sprint Schedule

Sprint 1 – Interface, Authentication, and System Setup (2 weeks)

May 25, 2026 - June 7, 2026

Sprint Goal

Develop the system foundation, including authentication, user management, Firebase setup, and role-based access control.

Selected Product Backlog Items

- Landing page
- About page
- Authentication Forms
- User Registration
- Firebase Authentication
- Role-Based Access Control
- Database Configuration

User Stories

User Story 1

As an administrator, I want a secure login system so that I can access administrative functions.

User Story 2

As a franchise operator, I require authentication credentials to access branch operations securely.

User Story 3

As a customer, I want to register for an account so that I can access appointment and monitoring features.

Sprint Backlog

- Configure React Project
- Configure Firebase Project
- Implement Authentication Pages
- Develop Registration Form
- Create Role-Based Login Logic

Development Tasks

- Install React dependencies
- Develop Landing Interface
- Develop About Interface
- Create Login Interfaces
- Develop Registration Interface
- Configure Firebase Authentication
- Implement Role-Based Routing
- Validate Authentication Inputs

Expected Increment

At the conclusion of Sprint 1, the system produces a working authentication module capable of handling administrator, operator, and customer login processes.

Sprint 2 – Client Module (2 weeks)

July 6, 2026 - July 21, 2026

Sprint Goal

Develop the customer-facing module, including AI chatbot services, appointment scheduling, service monitoring, and franchising applications.

Selected Product Backlog Items

- Customer Registration
- Customer Dashboard
- AI Chatbot
- Appointment Scheduling
- Service Monitoring Tracker
- Franchise Application Form

User Stories

User Story 1

As a customer, I want an AI chatbot so that I can receive information regarding services, requirements, and branch locations.

User Story 2

As a customer, I want appointment scheduling functionality so that I can arrange consultations with a selected branch.

User Story 3

As a customer, I want service tracking capabilities so that I can monitor the status of the services.

User Story 4

As a potential franchise applicant, I want to submit a franchise application so that I can apply for Fairfly franchising.

Sprint Backlog

- Create Customer Dashboard
- Develop Appointment Scheduling Module
- Integrate AI Chatbot
- Develop Service Monitoring Tracker
- Create Franchise Application Form

Development Tasks

- Customer Dashboard Interface
- Integrate Google Gemini API
- Configure Chatbot Responses
- Develop Appointment Scheduling Form
- Develop Franchise Application Form
- Connect Appointment Data to Firestore/Firebase
- Develop Service Status Tracker

Expected Increment

At the end of Sprint 4, customers can access inquiry services through the AI chatbot, schedule appointments, monitor service status, and submit franchising applications.

Sprint 3 – Head Office Admin Module (2 weeks)

June 22, 2026 - July 5, 2026

Sprint Goal

Develop the Head Office administrative module responsible for centralized monitoring, management, and configuration.

Selected Product Backlog Items

- Administrator Dashboard
- KPI Monitoring Dashboard
- Franchise Management
- Template Workflow Management
- Third-Party Links Management

User Stories

User Story 1

As an administrator, I want to monitor branch performance so that I can evaluate franchise productivity.

User Story 2

As an administrator, I want to create booking templates so that operators can follow standardized workflows.

User Story 3

As an administrator, I want to manage franchise accounts so that branch access can be controlled.

Sprint Backlog

- Develop Admin Dashboard UI
- Create KPI Dashboard
- Create Franchise Management Interface
- Create Workflow Management Module
- Implement Third-Party Link Module

Development Tasks

- Create Dashboard Components
- Create Services Management Interface
- Create Operators Management Interface
- Create Applications Management Interface
- Create History Management Interface
- Create Quick Links Management Interface
- Create a modal form for adding a new services
- Create a modal form for adding a new operator
- Create a modal form for adding a new link
- Configure Firebase for Team Chat
- Develop Franchise Management CRUD Functions
- Develop Workflow Template CRUD Functions

Expected Increment

At the end of Sprint 2, administrators can access a functional dashboard capable of monitoring performance, managing franchises, and configuring workflow templates.

Sprint 4 – Franchise Operator Module (2 weeks)

June 22, 2026 - July 5, 2026

Sprint Goal

Develop the Franchise Operator Module responsible for standardized booking workflows, inquiry handling, quotation generation, appointment management, and operator communication with the head office.

Selected Product Backlog Items

- Operator Dashboard

- Template-Driven Workflow
- Inquiry Form Management
- Quotation Form Management
- Appointment Management
- Team Chat Module
- History Management

User Stories

User Story 1

As a franchise operator, I want a guided booking workflow so that customer transactions follow standardized procedures.

User Story 2

As a franchise operator, I want to create inquiry and quotation forms so that client requests and service quotations are properly documented.

User Story 3

As a franchise operator, I want appointment management capabilities so that client schedules can be confirmed or cancelled.

User Story 4

As a franchise operator, I want to communicate with the administrator so that operational issues can be reported.

Sprint Backlog

- Create Operator Dashboard
- Develop Template Workflow Interface
- Develop Inquiry Form Module

- Develop Quotation Form Module
- Develop Appointment Management Module
- Implement Team Chat Module
- Develop History Page

Development Tasks

- Create Operator Dashboard Layout
- Load Workflow Templates from Firestore
- Develop Dynamic Form Components
- Implement Appointment Confirmation Logic
- Create Inquiry Submission Function
- Create Quotation Generation Form
- Implement Team Chat Messaging
- Store Transaction History

Expected Increment

At the conclusion of Sprint 3, franchise operators can process customer transactions using standardized booking workflows, manage inquiries and appointments, and communicate directly with the head office.

Sprint 5 – System Integration, Testing, Evaluation, and Deployment (2 weeks)

July 20, 2026 (Mon)August 2, 2026

Sprint Goal

Perform complete system integration, conduct testing procedures, evaluate system quality, and deploy the system.

Selected Product Backlog Items

- System Integration
- Black Box Testing
- ISO/IEC 25010 Evaluation
- Bug Fixing
- Deployment Configuration
- Firebase Hosting

User Stories

User Story 1

As a system administrator, I want all modules to operate seamlessly so that users experience a fully functional system.

User Story 2

As a researcher, I want to evaluate the system using ISO/IEC 25010 so that software quality standards are measured.

Sprint Backlog

- Integrate All Modules
- Conduct Black Box Testing
- Conduct ISO/IEC 25010 Evaluation
- Fix System Defects
- Configure Firebase Hosting
- Deploy Production Version
-

Development Tasks

- Integrate Admin, Operator, and Customer Modules
- Conduct Functional Testing
- Execute Black Box Test Cases
- Collect Evaluation Responses
- Calculate Weighted Mean Results
- Resolve Detected Bugs
- Configure Production Environment
- Deploy System

Expected Increment

At the end of Sprint 5, the researchers produce a fully integrated, tested, evaluated, and deployable web-based system.

Requirements Analysis and Documentation

The requirements for the creation and deployment of the developed Web-Based Centralized Real-Time Performance Monitoring and Integrated Booking Support System for Fairfly Franchises are outlined in this section. These requirements include the hardware and software specifications required for system development, as well as the operational specifications required for the system's successful deployment and operation. The system will function effectively, securely, and dependably while meeting the client's operational demands if these requirements are identified. The client was consulted, the current operational workflow was analyzed, and the technological resources required to support the developed system were assessed in order to identify the needs.

Hardware Requirements

The following hardware components are necessary to support the developed Web-Based Centralized Real-Time Monitoring and Integrated Booking Support System for Fairfly Franchises. Table 1 presents the minimum hardware specifications required for developing the system, including processor, memory, storage, and display requirements.

Table 1

Hardware Requirements for Development of the Proposed System

Hardware	Minimum Specification
Central Processing Unit (CPU)	Any Intel Core i3 processor or equivalent
Random Access Memory (RAM)	8 Gigabyte
Read-Only Memory (ROM) / Storage	128 Gigabyte
Graphics Processing Unit (GPU)	Any modern integrated graphics
Display	1920 x 1080 resolution

Software Requirements

The following software applications and frameworks are necessary to support the development of the developed Web-Based Centralized Real-Time Monitoring and Integrated Booking Support System for Fairfly Franchises. Table 2 presents the software applications and frameworks used in developing the system.

Table 2*Software Requirements for Development of the Proposed System*

Software	Description
Windows 10/11	Development will be performed mainly on the Windows operating system, which supports Node.js and other applications.
Visual Studio Code	An Integrated Development Environment (IDE) is used for coding and developing the web-based system.
React.Js	A JavaScript library that will be used to build the user interface of the web application.
Node.Js	JavaScript runtime environment used for running server-side scripts during development.
Node Package Manager (npm)	Package manager for installing and managing project libraries and dependencies
Firebase	A cloud-based platform that uses JSON as the real-time database and backend service for storing system data and managing authentication.
Git	A distributed version control system is used to track changes in source code during the software development of the proposed system
GitHub	Version control/Repository platform used for managing source code and tracking development progress.
Google Gemini	Powers the AI chatbot on the customer-facing landing page.
Web Browser	Used for system testing, debugging, and accessing the web-based platform.

Operational Requirements

The following requirements had to be met for the system to operate effectively upon deployment and throughout its long-term use. Table 3 specifies the operational requirements for deploying and maintaining the developed system, including database hosting, web hosting, internet connectivity, and role-based access control configurations.

Table 3*Operational Requirements upon Implementation*

Requirement	Description
Database	Firestore were used for real-time data storage and synchronization across all franchise branches and the head office.
Hosting	Firestore were used for deploying and serving the web application.
Internet	A stable and continuous internet connection is required at all times for all users.
Role-Based Access Control	The system must enforce role-based authentication, ensuring that administrators, franchise operators, and customers access only the features and data relevant to their roles.
Data Backup and Recovery	Regular automated backups of the Firestore database to prevent data loss. Firestore provides built-in backup and export capabilities that should be configured regularly.
System Maintenance	Periodic updates are a must for dependencies (React.js packages, Firestore SDK), security patches, and AI chatbot knowledge base updates to keep the system functional and secure.
Browser Compatibility	The system must be accessible and fully functional on the latest versions of all web browsers.
User Training	Franchise operators and head office administrators must be trained on how to use the system.

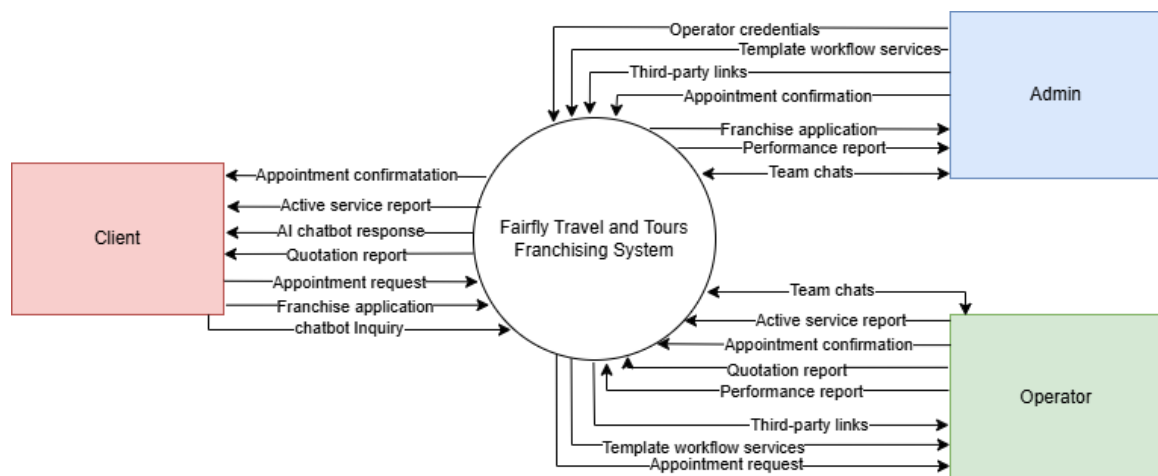
Design of Software, Systems, Product, and/or Processes**Context Diagram**

The Context Diagram provides an overview of the entire system. External entities interact with the core system through clearly labeled data flows. On the left, the rectangular entity labeled “Client” sends inputs including appointment request, franchise application, chatbot inquiry, quotation report, active service report, and appointment confirmation, while also receiving AI chatbot response and appointment confirmation. On the upper

right, the rectangular entity labeled “Admin” exchanges data such as operator credentials, template workflow services, third-party links, appointment confirmation, franchise application, performance report, and team chats. On the lower right, the rectangular entity labeled “Operator” exchanges team chats, active service report, appointment confirmation, quotation report, performance report, third-party links, template workflow services, and appointment request. All interactions are depicted with directional arrows showing the bidirectional flow of information between the external actors and the central system, clearly defining the system boundary. Figure 3 presents the context diagram of the developed system, identifying the external entities that interact with the system and the data flows between them.

Figure 3

Context Diagram of the Developed System

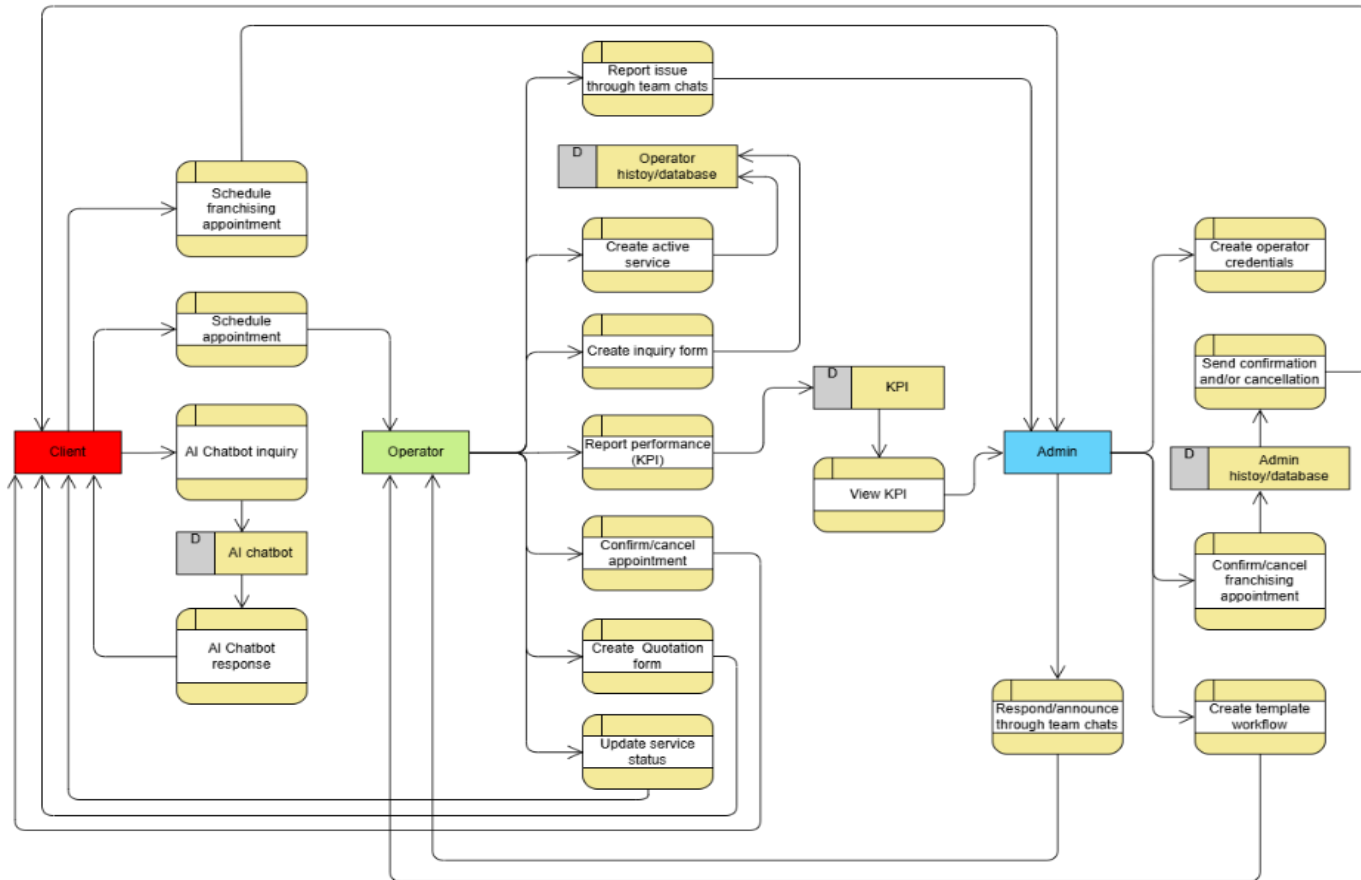


Level 1 Data Flow Diagram

Figure 4 illustrates the Level 1 Data Flow Diagram of the Fairly Travel and Tours Franchising System. External actors are represented as Client, Operator, and Admin. Data stores are denoted with “D” labels (AI chatbot, Operator history/database, KPI, and Admin history/database). The diagram decomposes the major processes into process boxes. The Client interacts with processes such as “Schedule franchising appointment,” “AI Chatbot inquiry,” and “AI Chatbot response.” The Operator performs processes including “Create active service,” “Create inquiry form,” “Create Quotation form,” “Confirm/cancel appointment,” “Update service status,” “Report performance (KPI),” and “Report issue through team chats.” The admin handles “Create operator credentials,” “Send confirmation and/or cancellation,” “Confirm/cancel franchising appointment,” “View KPI,” “Create template workflow,” and “Respond/announce through team chats.” Directional arrows show the complete internal data movement across providing a clear view of how information flows within the franchising platform.

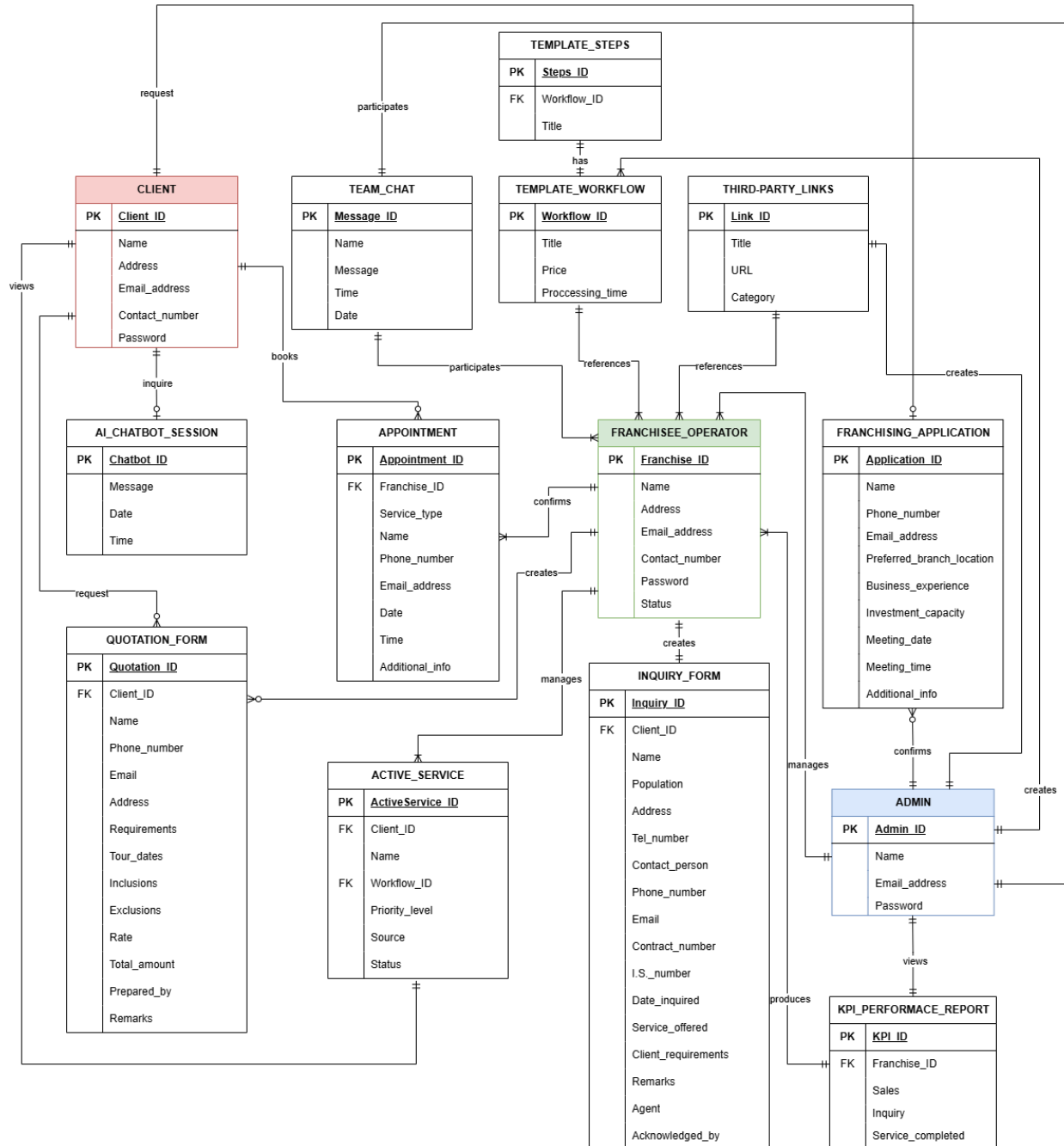
Figure 4

Level 1 Data Flow Diagram of the Developed System



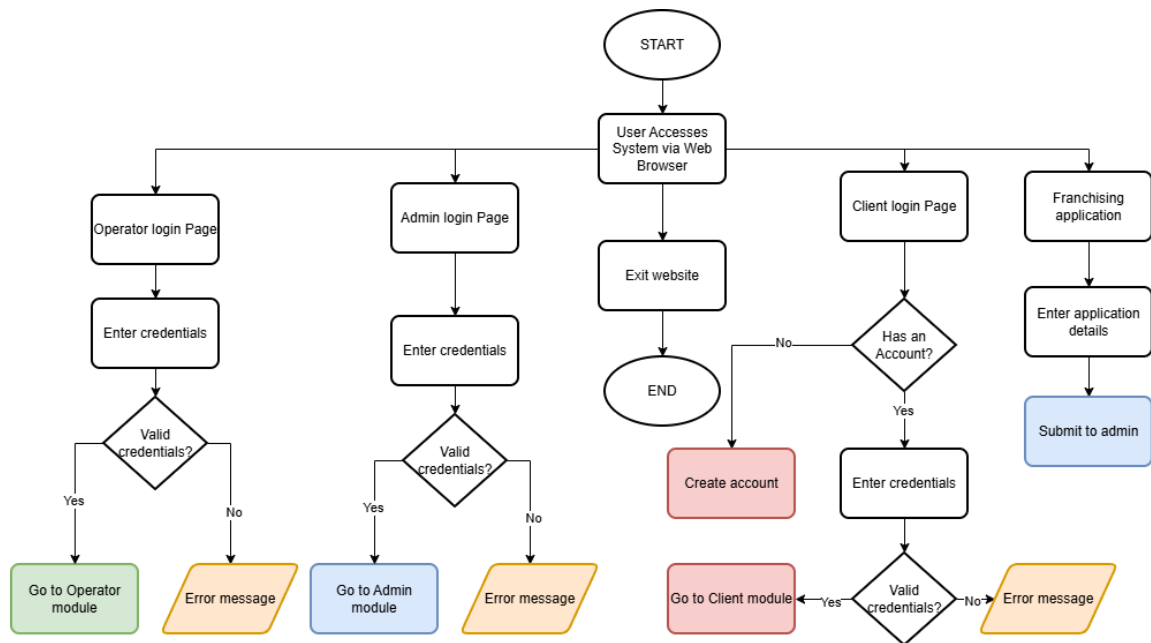
Enhanced Entity-Relationship Diagram

The Enhanced Entity-Relationship Diagram (EERD) illustrates the complete relational database schema for the Fairly Travel and Tours Franchising System. The diagram features multiple entities connected by labeled relationship lines that define the cardinality and interactions among entities. The central entities include CLIENT, FRANCHISEE_OPERATOR, and ADMIN. The CLIENT entity is associated with AI_CHATBOT_SESSION, QUOTATION_FORM, APPOINTMENT, and FRANCHISING_APPLICATION. The FRANCHISEE_OPERATOR entity creates INQUIRY_FORM, confirms APPOINTMENT, manages ACTIVE_SERVICE, creates INQUIRY_FORM, and participates in TEAM_CHAT. The ADMIN entity views KPI_PERFORMANCE_REPORT, creates TEMPLATE_STEPS and TEMPLATE_WORKFLOW, and manages various components. Additional supporting tables include TEAM_CHAT, TEMPLATE_WORKFLOW, THIRD-PARTY_LINKS, TEMPLATE_STEPS, and ACTIVE_SERVICE, each linked through explicit relationships such as “has,” “references,” “participates,” “confirms,” “manages,” and “views.” All relationships are clearly labeled, ensuring full traceability of data dependencies across the franchising platform, including chatbot sessions, appointment booking, quotation and inquiry forms, service monitoring, and performance reporting. Figure 5 presents the EERD of the developed system, defining the database entities, their attributes, and the relationships that govern data dependencies across the franchise management, booking, and user authentication components.

Figure 5*EERD of the Developed System*

Flowcharts

The Landing page Flowchart details the initial entry points and authentication logic for all user types in the Fairfly system. It starts at a white oval labeled “START,” followed by “User Accesses System via Web Browser.” From there, four parallel paths emerge: Operator Login Page, Admin Login Page, Client Login Page, and Franchising Application. Each login path requires the user to “Enter credentials,” followed by a diamond decision “Valid credentials?” A “Yes” response routes the user to the appropriate module. A “No” response triggers an orange “Error message” and returns the user to the login screen. For new clients, a diamond “Has an Account?” leads to “Create account” if the answer is “No,” or to credential entry if “Yes.” The Franchising Application path requires the user to “Enter application details” and “Submit to admin.” The flowchart concludes at a white oval labeled “END” or “Exit website.” This diagram provides a comprehensive view of role-based access control.

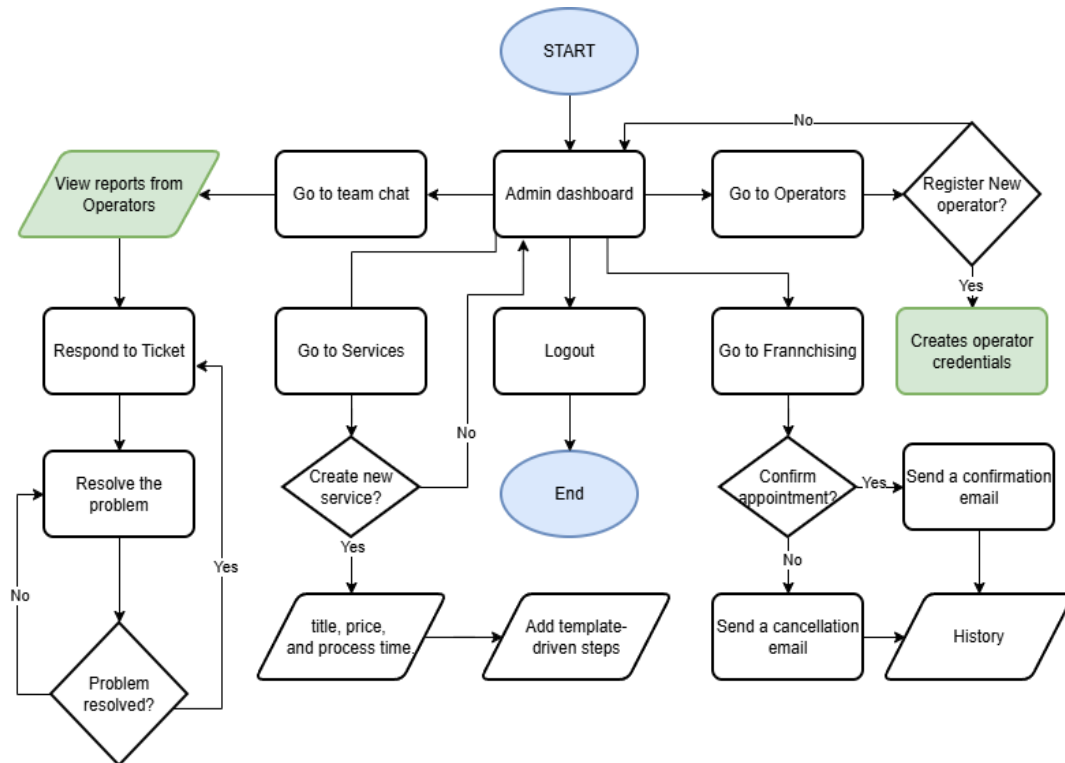
Figure 6*Landing Page Flow Chart*

The Admin Flowchart depicts the complete workflow within the Admin Dashboard of the Fairfly system. It begins at an oval labeled “START” and immediately presents the “Admin dashboard.” From this central point, the admin can select multiple functions: “View reports from Operators,” “Go to team chat,” “Go to Operators,” “Go to Services,” “Go to Franchising,” “Logout,” and “Register New operator?” If registering a new operator, the system creates operator credentials. The “Respond to Ticket” path leads to “Resolve the problem,” followed by a diamond “Problem resolved?” A “Yes” response returns to the dashboard, while “No” loops back for further action. The “Create new service?” decision node, when answered “Yes,” requires input of “title, price, and process time” and “Add template-driven steps.” The “Confirm appointment?” decision routes to

“Send a confirmation email” (Yes) or “Send a cancellation email” and “History” (No). The flowchart terminates at a blue oval labeled “End.”

Figure 7

Flow Chart Diagram of the Admin Module

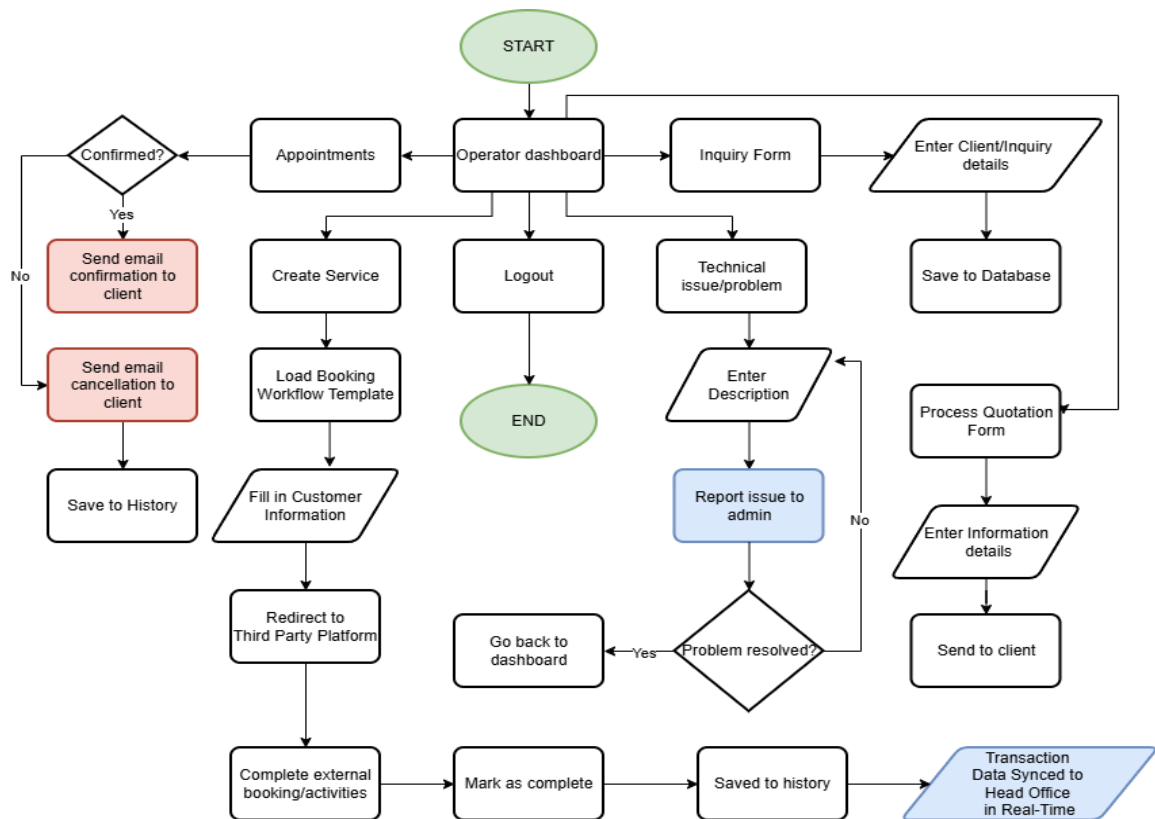


The Operator Flowchart outlines the operational workflow for franchise operators in the Fairfly system. It starts at an oval labeled “START” and proceeds directly to the “Operator dashboard.” Key functions include “Appointments,” “Inquiry Form,” “Technical issue/problem,” “Enter Client/Inquiry details,” “Create Service,” “Load Booking Workflow Template,” and “Logout.” Upon receiving an inquiry, the operator enters a description, saves it to the database, and processes the quotation form. A diamond decision “Confirmed?” determines whether to “Send email confirmation to client” or “Send

email cancellation to client”. The booking path involves “Fill in Customer Information,” “Load Booking Workflow Template,” possible redirection to a “Third Party Platform,” “Complete external booking/activities,” “Mark as complete,” and “Saved to history.” If a problem arises, the operator can “Report issue to admin” or check “Problem resolved?” with appropriate branching. All transactions conclude with “Transaction Data Synced to Head Office”. The flowchart ends at an oval labeled “END.”

Figure 8

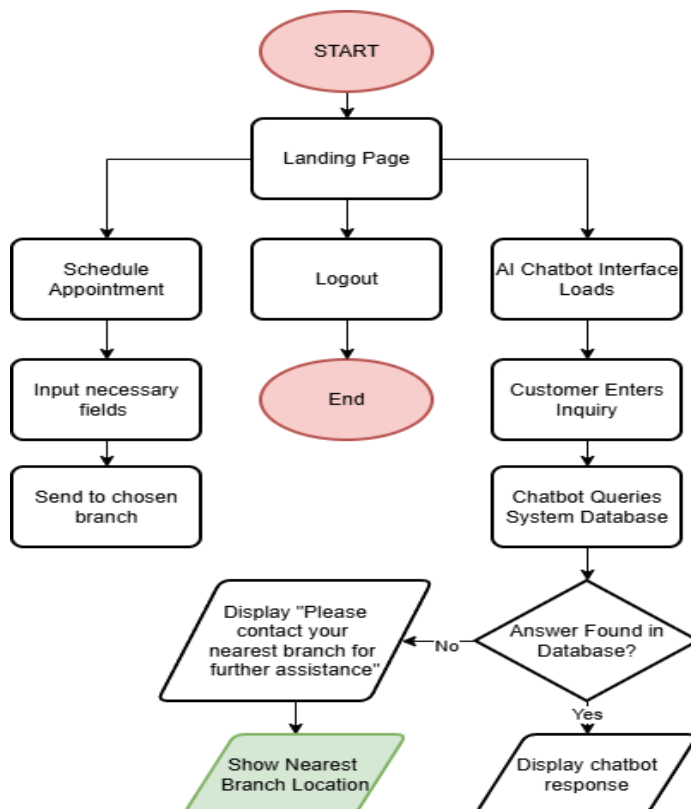
Flow Chart Diagram of the Operator Module



The Client Flowchart illustrates the end-to-end journey of a client interacting with the Fairfly web-based Centralized Franchise System. It begins at the Landing Page, and then branches into three primary options: Schedule Appointment, Logout, and AI Chatbot Interface. Upon selecting the chatbot path, the customer enters an inquiry, after which the system executes the Chatbot Queries System Database. A diamond decision node checks “Answer Found in Database?” If the answer is “Yes,” the system displays the chatbot response. If the answer is “No,” it displays the message “Please contact your nearest branch for further assistance” and shows the nearest branch location. The “schedule appointment” path requires the customer to input the necessary fields before sending the request to the chosen branch. The flowchart terminates at an oval labeled “End.”

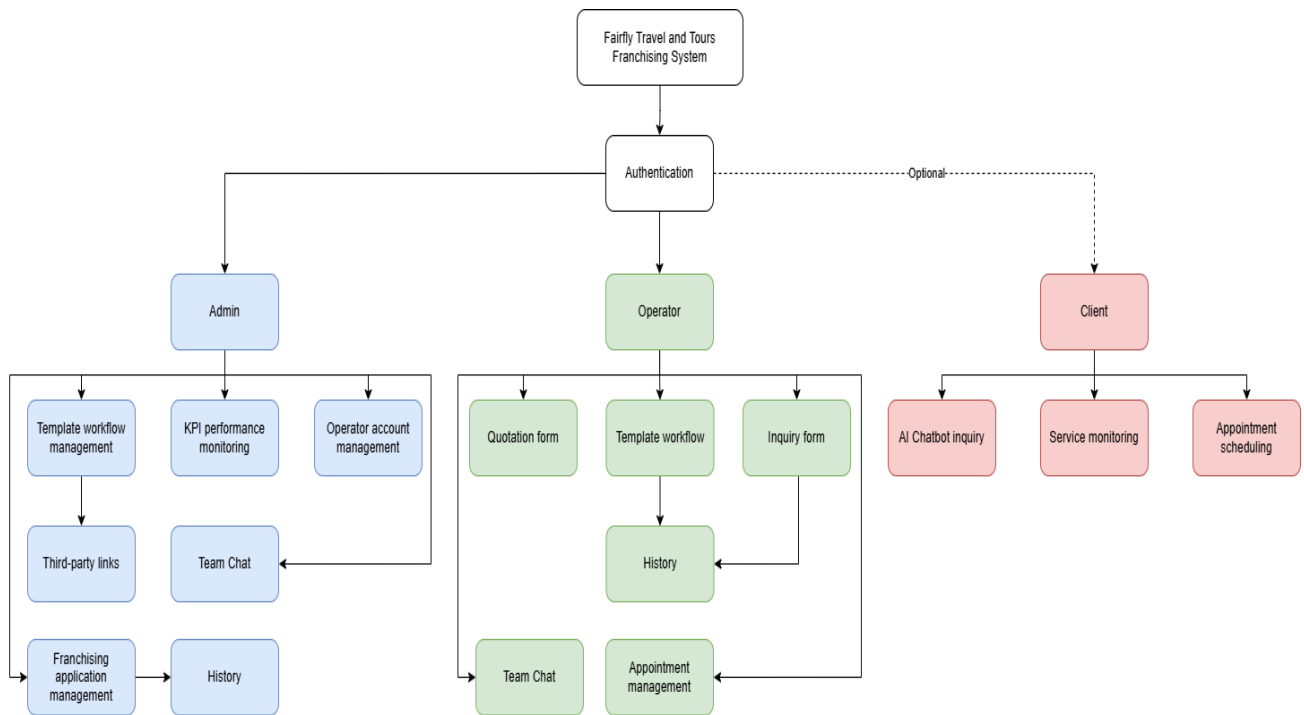
Figure 9

Flow Chart Diagram of the Client Module



Visual Table of Contents

The Visual Table of Contents presents a hierarchical overview of the modular architecture of the Franchising System. At the top, a rectangular box labeled “Fairly Travel and Tours Franchising System” connects downward to the “Authentication” module. Below the Authentication, the diagram branches into three primary role-based sections. The left column under “Admin” lists Template workflow management, KPI performance monitoring, Operator account management, Third-party links, Team Chat, Franchising application management, and History. The middle column under “Operator” contains Quotation form, Template workflow, Inquiry form, History, Team Chat, and Appointment management. The right column under “Client” includes AI Chatbot inquiry, Service monitoring, and Appointment scheduling. This tree-like structure visually organizes the system’s features by user role, providing a clear, modular breakdown of the entire franchising platform for quick reference. Figure 10 presents the Visual Table of Contents (VTOC) diagram, summarizing the system's modular architecture by organizing the Head Office, Franchise Operator, and Customer modules into a hierarchical structure.

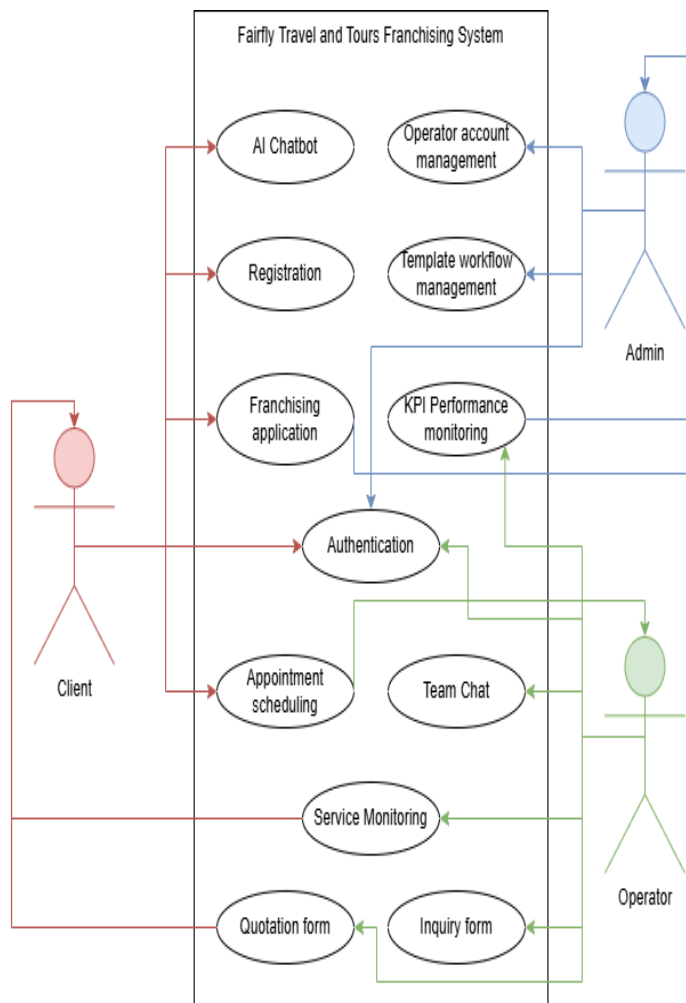
Figure 10*Visual Table of Contents of the Developed System***Use Case**

The Use Case Diagram illustrates the functional requirements and actor interactions within the Fairfly Travel and Tours Franchising System. Three actors are represented by stick figures: the Client, the Operator, and the Admin. The system boundary contains multiple oval use cases connected by solid and dashed arrows indicating associations and relationships. The Client actor interacts with the AI Chatbot, Registration, Franchising application, Appointment scheduling, Quotation form, and Inquiry form. The Operator actor connects to Operator account management, Template workflow management, Team Chat, Service Monitoring, Inquiry form, Quotation form, and Appointment management. The Admin actor links to Operator account management, Template workflow management,

KPI Performance monitoring, Team Chat, and Authentication. Authentication serves as a central use case connected to most others, while an “Optional” note appears beside the Authentication use case. The diagram clearly shows role-based access and the extend/include relationships among the system functionalities. Figure 11 presents the use case diagram of the developed system, identifying the primary actors and the key system functions each actor can perform, including franchise management, booking template operations, performance monitoring, support ticketing, and AI chatbot inquiries.

Figure 11

Use Case Diagram of the Developed System



Testing

This section presents the research instruments, respondents, and statistical treatment that were used in evaluating the developed system.

Research Instrument

The developed system was evaluated using a structured questionnaire based on the ISO/IEC 25010:2023 software product quality model. The ISO/IEC 25010 is an internationally recognized standard that defines the set of quality characteristics for evaluating software systems (International Organization for Standardization [ISO], 2023).

The following quality characteristics are selected as the evaluation criteria for the developed system:

1. Functional Suitability - This characteristic assesses whether the system's features, including booking template management, franchise management, and dashboard monitoring, perform their intended functions correctly (ISO, 2023).
2. Performance Efficiency - This characteristic evaluates whether the system responds, processes transactions, and loads data within acceptable timeframes, particularly the real-time dashboard updates and booking workflow operations (ISO, 2023).
3. Usability - This characteristic evaluates whether the system's interface is easy to navigate, intuitive to learn, and provides a satisfactory user experience for administrators and franchise operators (ISO, 2023).
4. Reliability - This characteristic evaluates whether the system operates consistently without failures, errors, or unexpected interruptions during the use of the system (ISO, 2023)

In addition to the ISO/IEC 25010 evaluation, Black-box Testing was also conducted as a mandatory component of the system testing process. Black-box Testing is a software testing technique that tests the system without looking at or examining its internal code structure. A tester provides an input and observes as the system being tested provides an output. This allows for an evaluation of the system's response time, usability and reliability issues, and how it reacts to both expected and unexpected user activities. (Jameson, 2023)

Respondent of the Study

The respondents of this study consisted of the Fairfly Travel and Tours management and employees who will serve as the primary end-users of the developed system, as well as IT Experts who will provide a technical assessment of the system's design and functionality. Table 4 presents the distribution of these respondents, categorized by their roles within the evaluation process.

Table 4

Distribution of Respondents

Respondents Category	Frequency (f)	Percentage (%)
Owner/Management	2	28.57% - 22.22%
Employee	2	28.57% - 22.22%
IT Expert	3-5	42.86% - 55.56%
Total	7-9	100%

The Owner/Management personnel were included as respondents because they will operate the Head Office (Administrator) Module and have direct custody of its franchise operations. The employees were included because they represent the future franchise

operators who will use the Franchise Operator Module, including the template-driven booking workflow.

IT Experts were also included to provide an objective technical evaluation of the developed system's interface design, functionality, and overall system architecture. Their expertise in software development and web-based systems enables them to assess from a technical perspective, helping the end-user evaluation by Fairfly's management and employees.

The researchers used purposive sampling, a non-probability sampling technique in which the respondents are selected based on specific criteria relevant to the study (Nikolopoulou, 2022). This technique was chosen because the developed system is mainly a B2B platform designed specifically for Fairfly Travel and Tours. The end-user respondents (management and employees) were selected as the actual intended users of the system, while the IT Experts were selected based on their professional background and technical competence in evaluating web-based applications. This combination ensures that the prototype is assessed from both an operational and technical standpoint.

Inclusion Criteria:

For Owner/Management and Employees:

- Currently employed or affiliated with Fairfly Travel and Tours
- Directly involved in the agency's booking operations, management, or administrative processes
- Willing to participate in the prototype evaluation voluntarily

For IT Experts:

- Possesses a degree or professional background in Information Technology, Computer Science, or a related field
- Has experience in web-based system development or software evaluation
- Willing to participate in the prototype evaluation voluntarily

Exclusion Criteria:

- Individuals not affiliated with Fairfly Travel and Tours (for end-user respondents)
- Individuals with no professional background or experience in IT (for IT Expert respondents)

Statistical Treatment

The data gathered from the evaluation questionnaire were analyzed using descriptive statistics, specifically frequency, percentage, and weighted mean. Frequency and percentage were used to describe the demographic profile of the respondents, while the weighted mean was used to determine the overall rating of the system across each ISO/IEC 25010 quality characteristic. To interpret these results, the data was quantified using the five-point Likert scale presented in Table 5.

Table 5

Five-point Likert Scale

Scale	Verbal Interpretation	Range
5	Excellent	4.21 – 5.00
4	Very Good	3.41 – 4.20
3	Good	2.61 – 3.40
2	Fair	1.81 – 2.60
1	Poor	1.00 – 1.80

Description of the Prototype

The prototype developed for this study is a Web-Based Centralized Real-Time Performance Monitoring and Integrated Booking Support System designed for Fairfly Travel and Tours Franchises. It was created using Figma, a collaborative web-based design tool for user interface (UI) and user experience (UX) design, to provide a high-fidelity visual representation of the developed system before proceeding to full development (*What Is Figma Used For?*, n.d.).

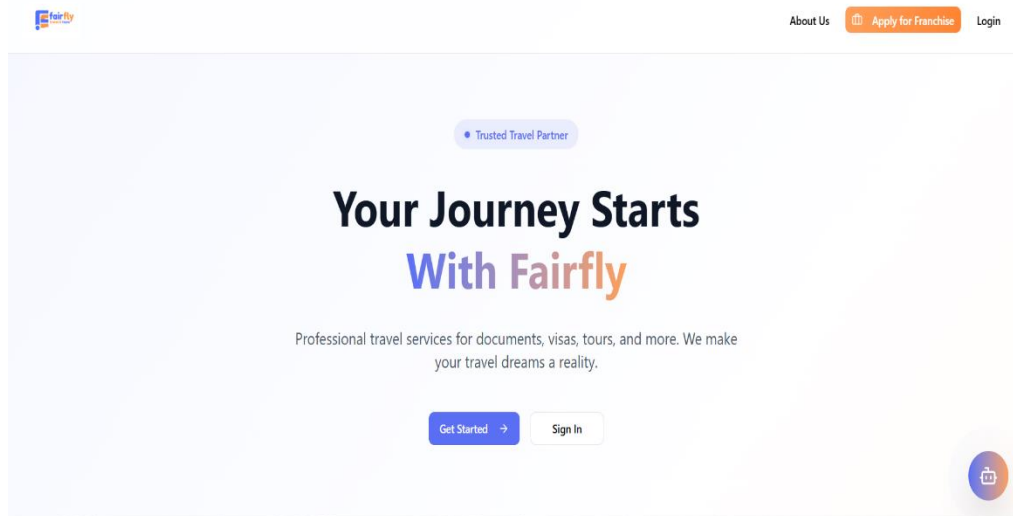
The prototype addresses the operational challenges that Fairfly currently faces as it transitions from a single-branch, office-based agency to a franchise business model. Currently, the agency relies on manual processes for booking transactions, record-keeping, and communication, which limits its ability to scale operations, maintain service consistency, and monitor franchise branch performance. The prototype demonstrates how the developed system solved these issues through a centralized online platform that connects the head office, franchise operators, and customers under a unified environment.

The prototype is structured around three (3) main modules. The Head Office (Administrator) Module provides the administrator with tools for franchise management, real-time performance monitoring through a centralized dashboard, franchise account management, booking template creation and deployment, resource library management, and a technical support ticketing system. The Franchise Operator Module allows branch operators to process customer transactions through a template-driven booking workflow, a guided, step-by-step process created by the administrator that standardizes how bookings are handled across all branches. When a booking step requires interaction with an external provider, the operator is redirected to the designated third-party platform and returns to the

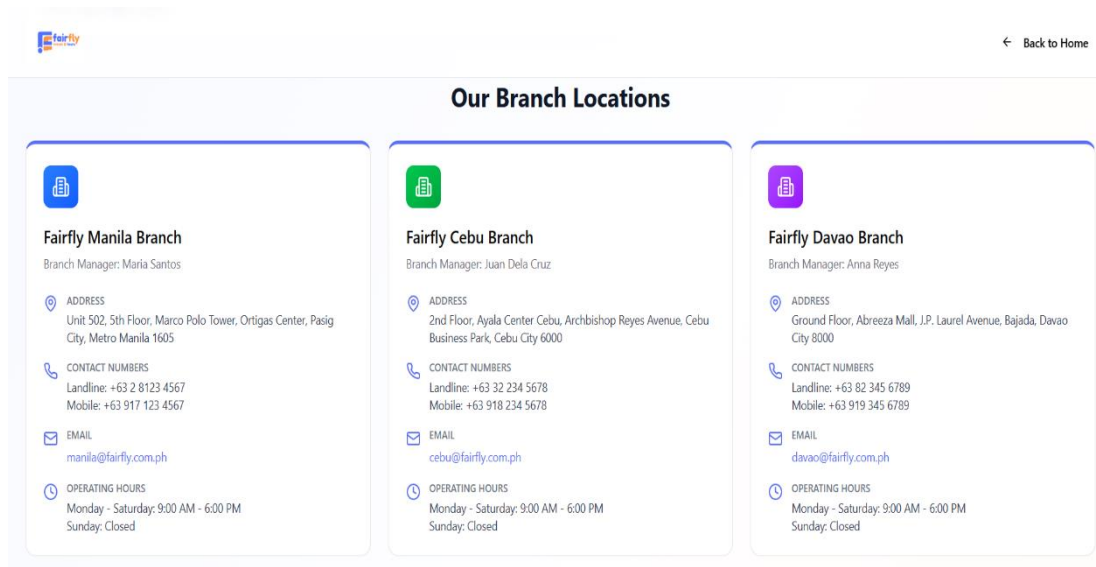
system to record the transaction details before proceeding to the next step. The Customer Module features a public-facing inquiry interface with an AI-powered chatbot that draws its responses from the system's database, providing potential clients with information about services, requirements, and branch locations.

The design approach emphasizes clarity, consistency, and role-based navigation. Each user role is directed to a distinct interface tailored to their specific functions, ensuring that administrators, operators, and customers access only the features relevant to their roles. The prototype incorporates interactive elements such as clickable buttons, page transitions, and form interactions to simulate the actual user experience during consultations with Fairfly management.

What distinguishes this prototype from existing travel agency systems is its template-driven booking workflow, which standardizes franchise operations without requiring direct API integration with any third-party booking platforms. Rather than building a custom booking system, the system enables franchise operators to process transactions on behalf of customers using admin-controlled templates, ensuring procedural consistency across all branches while capturing complete transaction data for centralized monitoring. This approach allows Fairfly to scale its franchise operations without the complexity and cost of integrating with multiple external booking APIs. Figures 12 to 28 provide a visual walkthrough of the Fairfly prototype, highlighting the key screens designed to standardize franchise transactions and centralize head-office oversight.

Figure 12*Landing Page of the Fairfly System*

The main landing page introduces Fairfly Travel and Tours' services and provides navigation to login, registration, franchise application, franchise list, and customer inquiry.

Figure 13*Franchise List Page*

A page displaying the list of active Fairfly franchise branches and their locations.

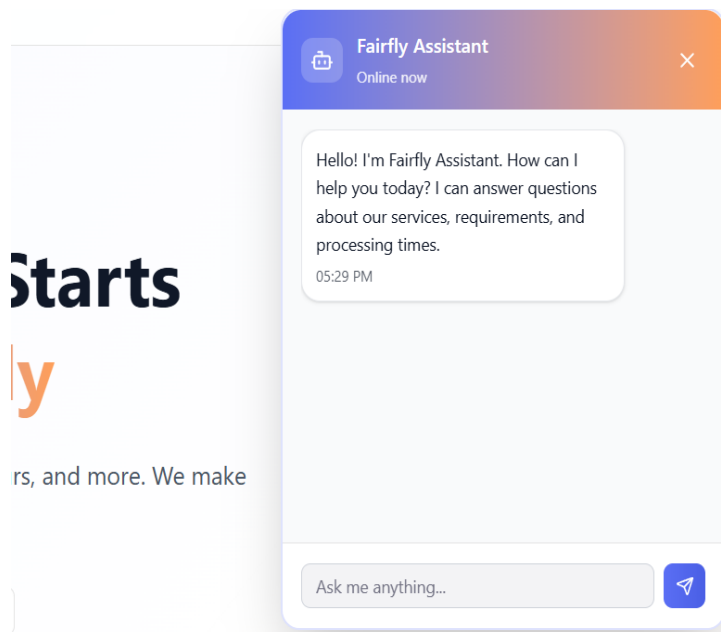
Figure 14*Franchise Application Page*

The screenshot displays the Fairfly Travel Agency UI. A central modal window titled "Apply for Fairfly Franchise" is open, allowing users to submit an application. The form includes the following fields:

- Full Name ***: Juan Dela Cruz
- Phone Number ***: +63 912 345 6789
- Email Address ***: your.email@example.com
- Preferred Branch Location ***: e.g., Quezon City, Makati, Cebu City
- Business Experience ***: Select your experience level (dropdown menu)
- Investment Capacity ***: Select investment range (dropdown menu)
- Preferred Meeting Date *** and **Preferred Meeting Time *** (dropdown menus)

The background of the UI shows various service categories: PSA Birth Certificate, Package Tours, PSA Assistance, and Document Services. Navigation links for "About Us", "Apply for Franchise", and "Login" are visible in the top right corner.

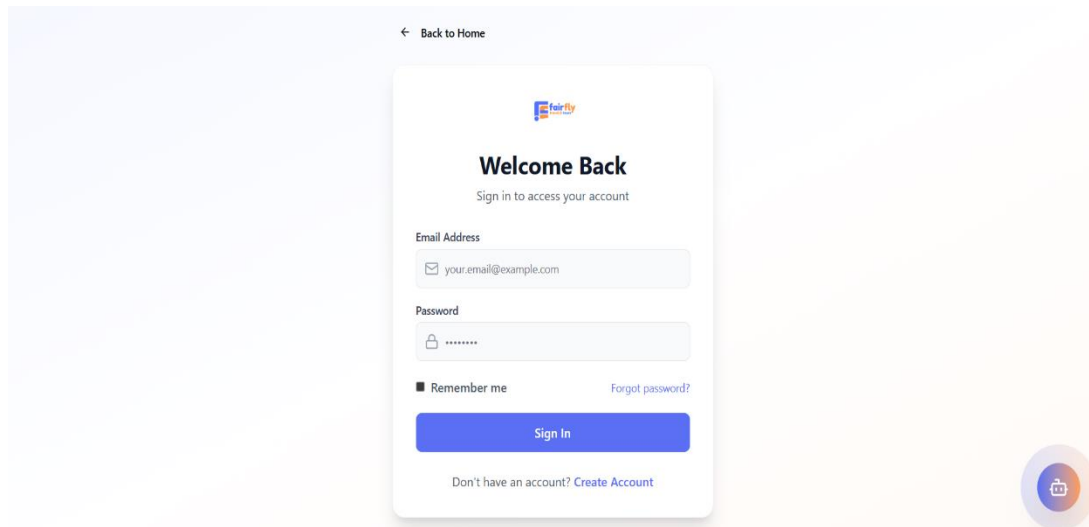
A public-facing form where potential franchisees can submit an application to operate a Fairfly Travel and Tours branch.

Figure 15*Customer Chatbot Interface*

The AI chatbot interface is where users can interact with the system to inquire about Fairfly's services, requirements, and branch locations.

Figure 16

Client Login Page

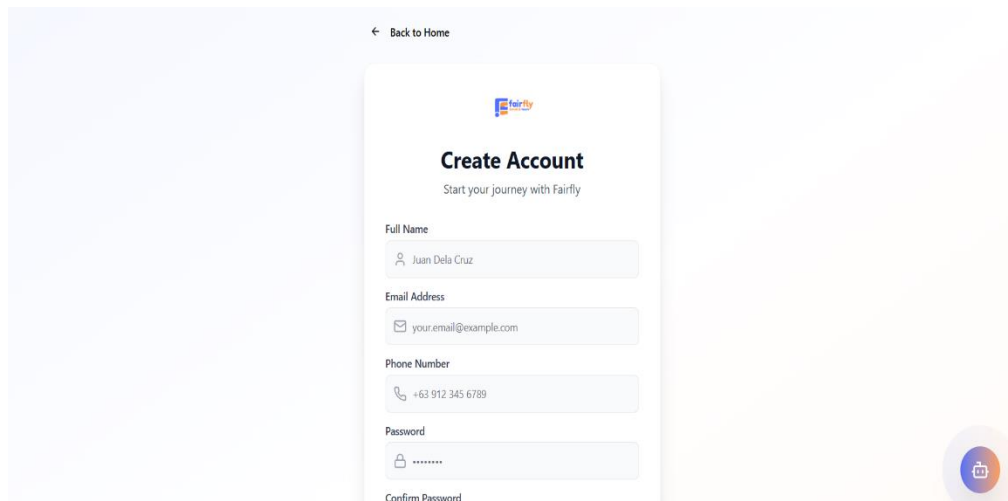


The image shows a mobile app interface for the 'Client Login Page'. At the top left, there is a back arrow and the text 'Back to Home'. The main content area features a white card with the 'fairfly' logo at the top. Below the logo, the text 'Welcome Back' is displayed in bold, followed by 'Sign in to access your account'. The card contains two input fields: 'Email Address' with a placeholder 'your.email@example.com' and 'Password' with a placeholder '*****'. Below these fields are two links: 'Remember me' with a checked checkbox and 'Forgot password?'. A blue 'Sign In' button is positioned below the links. At the bottom of the card, there is a link 'Don't have an account? Create Account'. In the bottom right corner of the screen, there is a circular icon with a shopping cart symbol.

This authentication page is where clients, operators, and the admin enter their credentials to access their account.

Figure 17

Client Registration Page

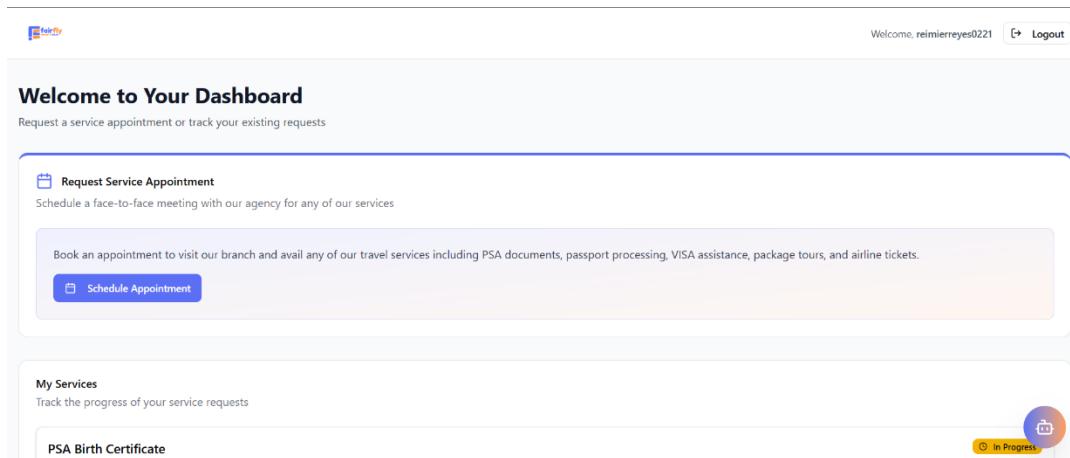


The image shows a mobile app interface for the 'Client Registration Page'. At the top left, there is a back arrow and the text 'Back to Home'. The main content area features a white card with the 'fairfly' logo at the top. Below the logo, the text 'Create Account' is displayed in bold, followed by 'Start your journey with Fairfly'. The card contains four input fields: 'Full Name' with a placeholder 'Juan Dela Cruz', 'Email Address' with a placeholder 'your.email@example.com', 'Phone Number' with a placeholder '+63 912 345 6789', and 'Password' with a placeholder '*****'. Below the password field, there is a label 'Confirm Password'. In the bottom right corner of the screen, there is a circular icon with a shopping cart symbol.

An optional registration form that allows clients to create an account for accessing the system's monitoring and appointment feature.

Figure 18

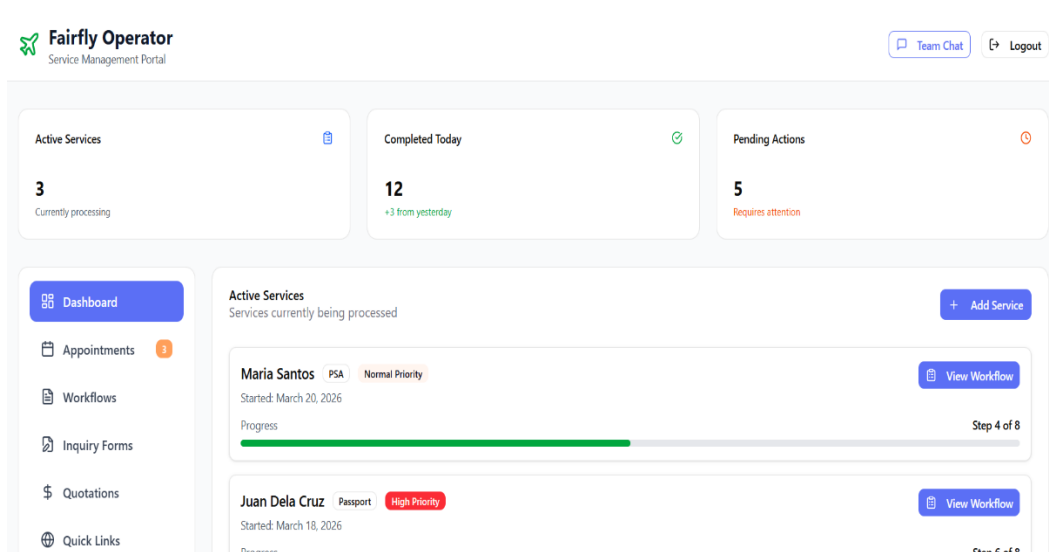
Client Dashboard



A dashboard that allows clients to schedule an appointment to their chosen branch and monitor their availed services.

Figure 19

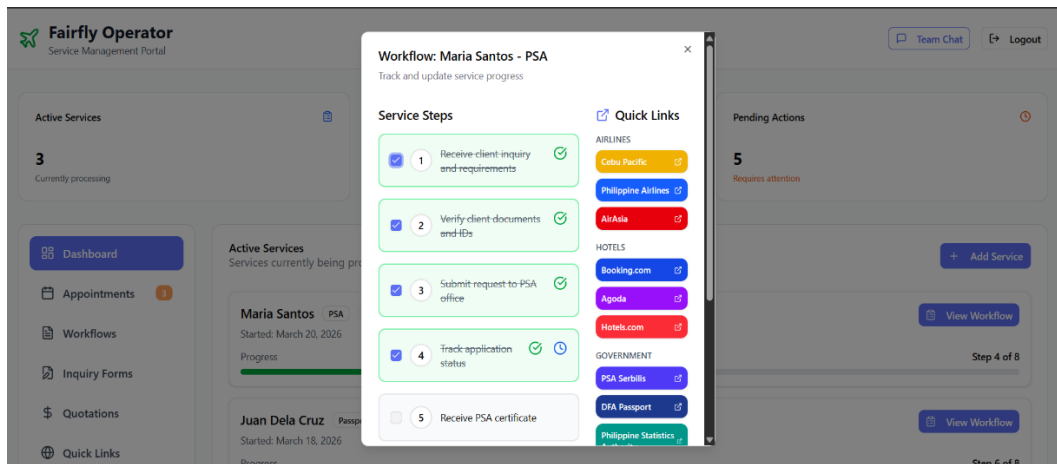
Franchise Operator Dashboard



The main dashboard for franchise operators, displaying branch-level booking activity, pending transactions, and notifications from the head office.

Figure 20

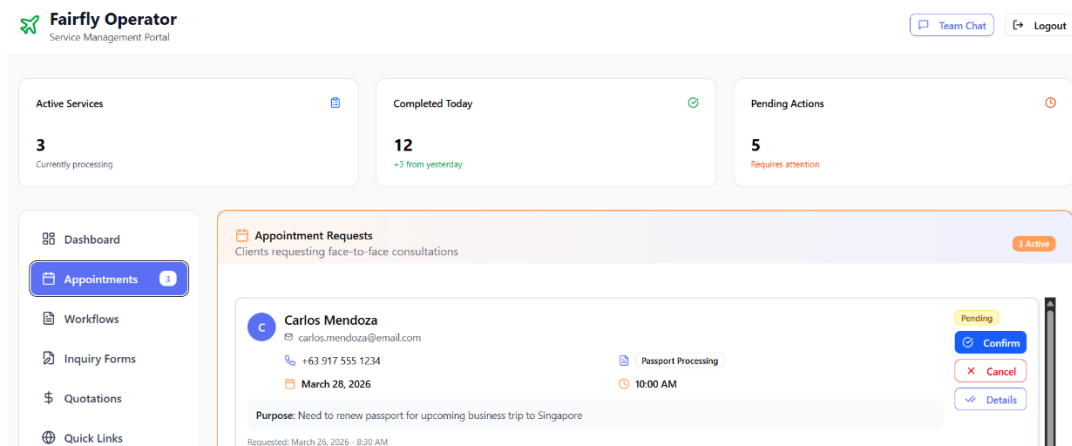
Template Workflow Page



The step-by-step booking template interface, where operators process customer transactions by completing required fields, with redirection to designated third-party booking platforms at applicable steps.

Figure 21

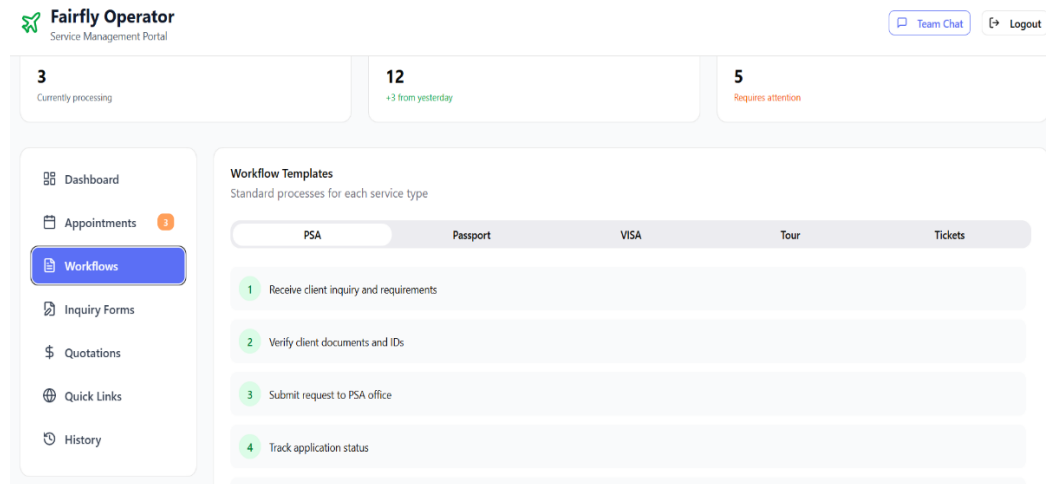
Appointment Management Page



The dashboard for displaying pending appointment request from potential clients. This is where operator can accept or cancel appointment request.

Figure 22

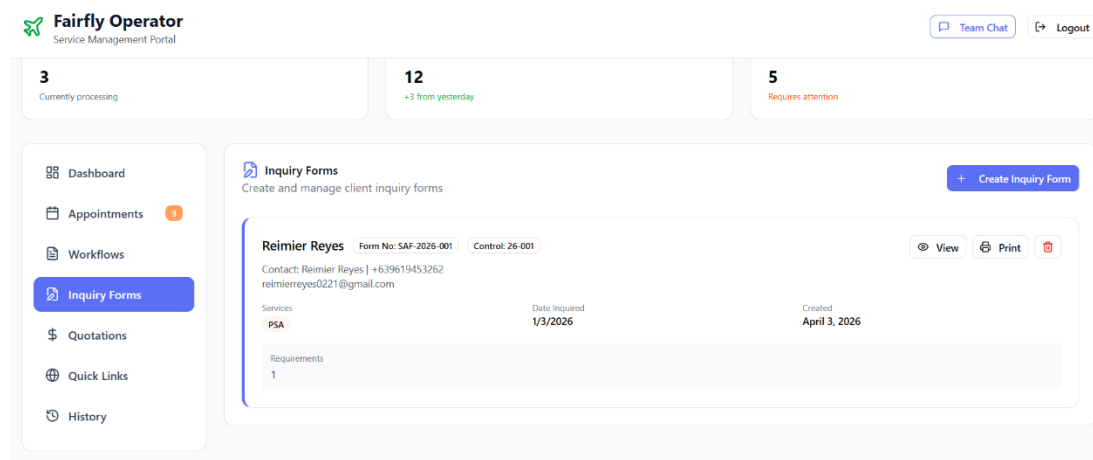
Workflow Page



The operator's operations interface displays all available templates created by the administrator, organized by service type.

Figure 23

Inquiry Form Management Page



The Inquiry form interface can create and displays all inquiries from current and potential client.

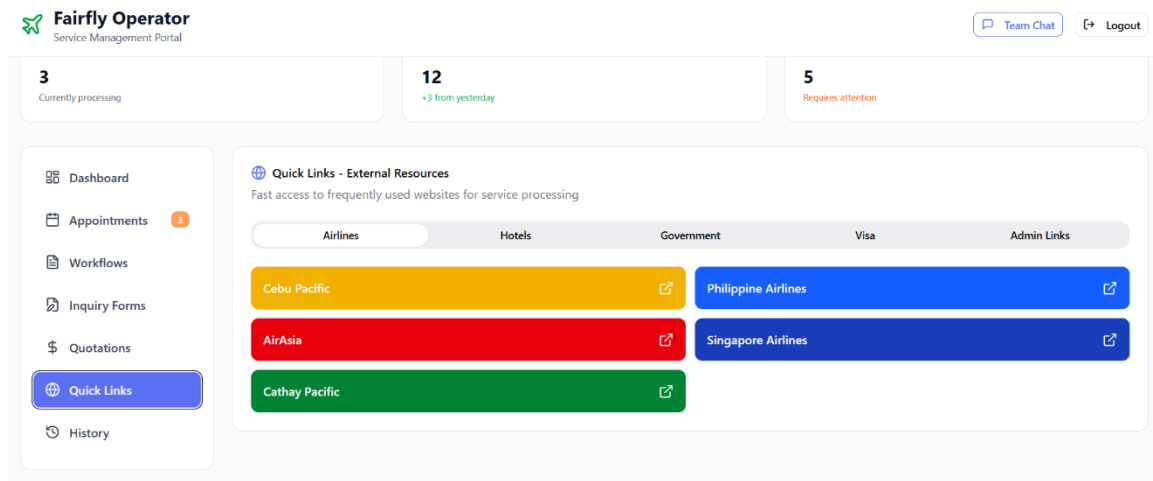
Figure 24

Quotation Form Management Page

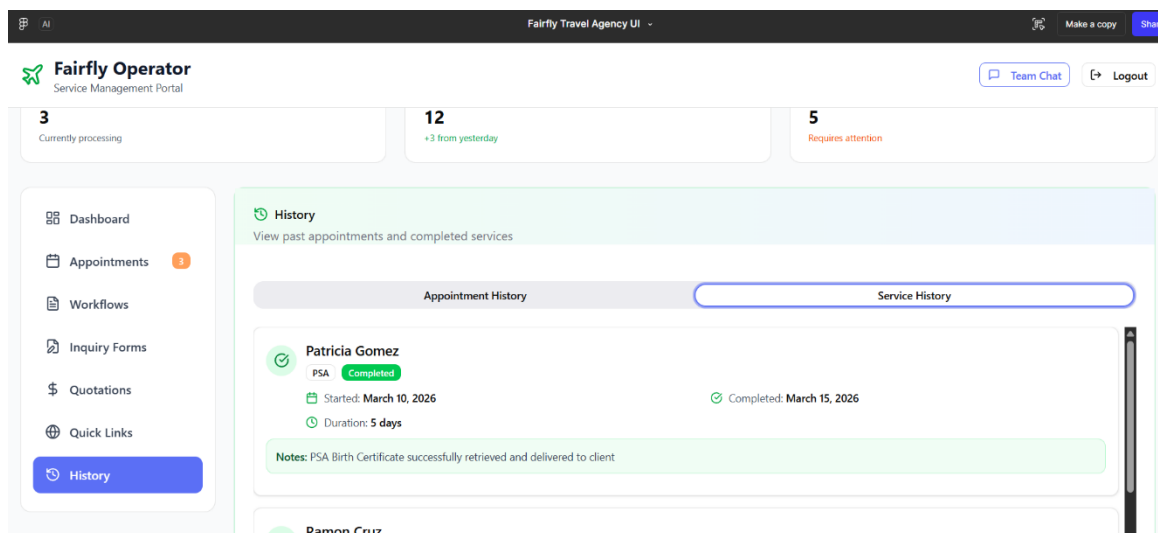
The screenshot displays the Fairfly Operator Service Management Portal. At the top, the logo 'Fairfly Operator' and 'Service Management Portal' are visible on the left, and 'Team Chat' and 'Logout' buttons are on the right. Below the header, there are three summary cards: '3 Currently processing', '12 +3 from yesterday', and '5 Requires attention'. The main content area is titled 'Quotation Forms' with the subtitle 'Create and manage tour quotations'. A '+ Create Quotation' button is in the top right. A sidebar on the left lists navigation options: Dashboard, Appointments (with a red notification badge), Workflows, Inquiry Forms, Quotations (highlighted in blue), Quick Links, and History. The main content area shows a form for a quotation for 'Reimier Reyes' with ID 'QT-2026-0001' and status 'Draft'. The form includes fields for contact information (+639619453262, reimierreyes0221@gmail.com), tour dates, and preparation details. It also displays a table with columns for 'Rate (per person)', 'Total Amount', and 'Date Created'. The table shows a rate of ₱1.00, a total amount of ₱1.00, and a date created of 4/3/2026. A 'Mark as Sent' button is located at the bottom of the form.

Rate (per person)	Total Amount	Date Created
₱1.00	₱1.00	4/3/2026

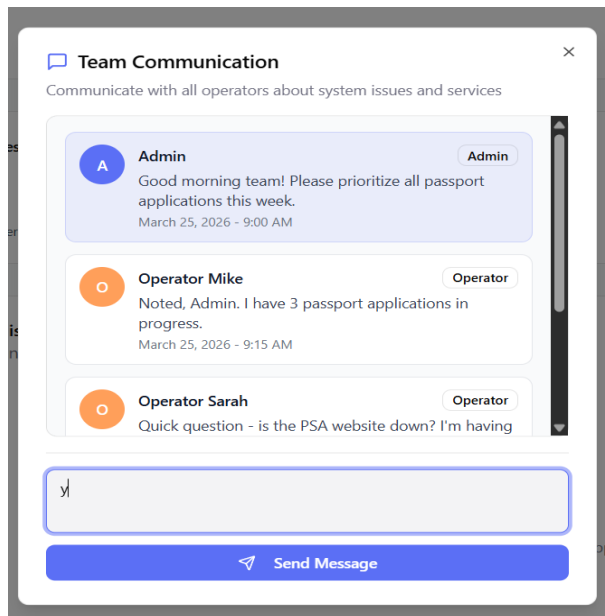
The Quotation form interface can create and displays all quotation forms from current client that availed Fairfly services.

Figure 25*Quick Link Page*

The Quick link page displays all links of third-party websites that is used in completing services.

Figure 26*Operator History Page*

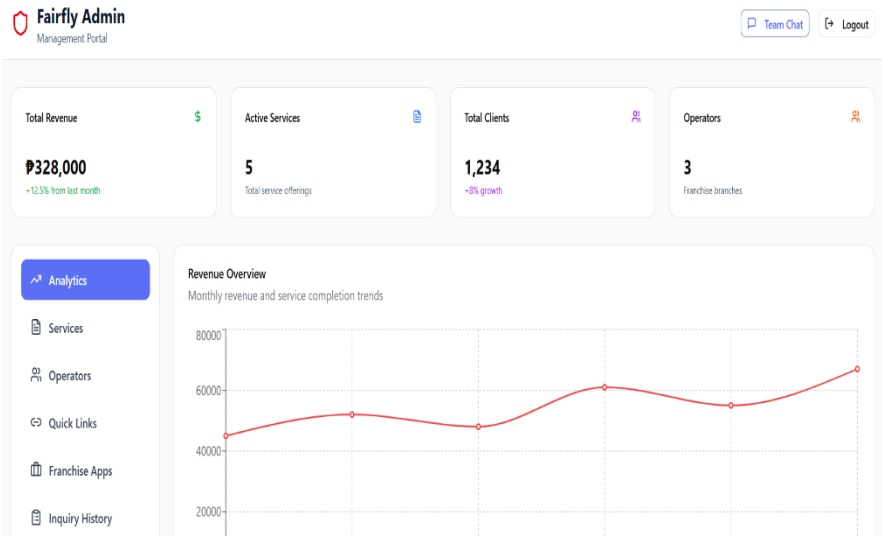
The History page displays all history of services and appointment completed or cancelled.

Figure 27*Team Chat*

The interface where franchise operators submit messages about technical or operational problems directly to the head office. Can also be used by admin for announcements.

Figure 28

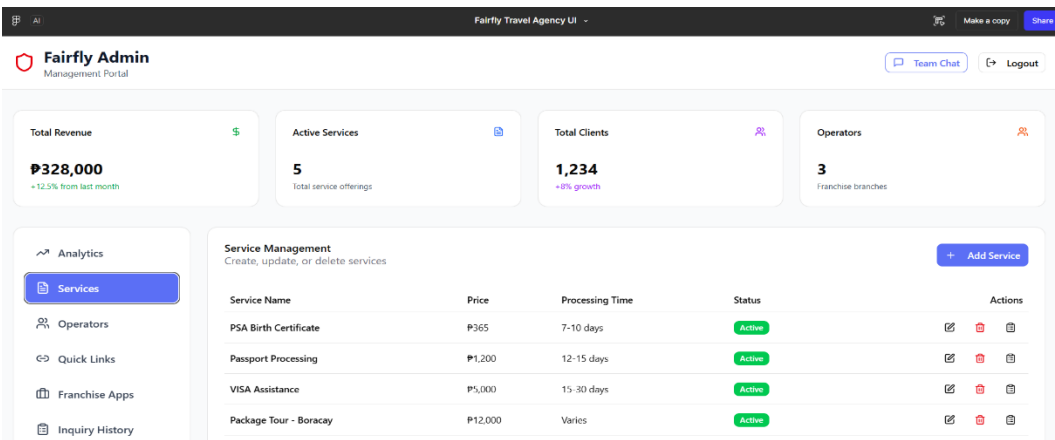
Administrator Dashboard



The centralized dashboard for the head office, displays real-time Key Performance Indicators (KPIs) such as total sales, booking volume, and franchise branch status across all locations.

Figure 29

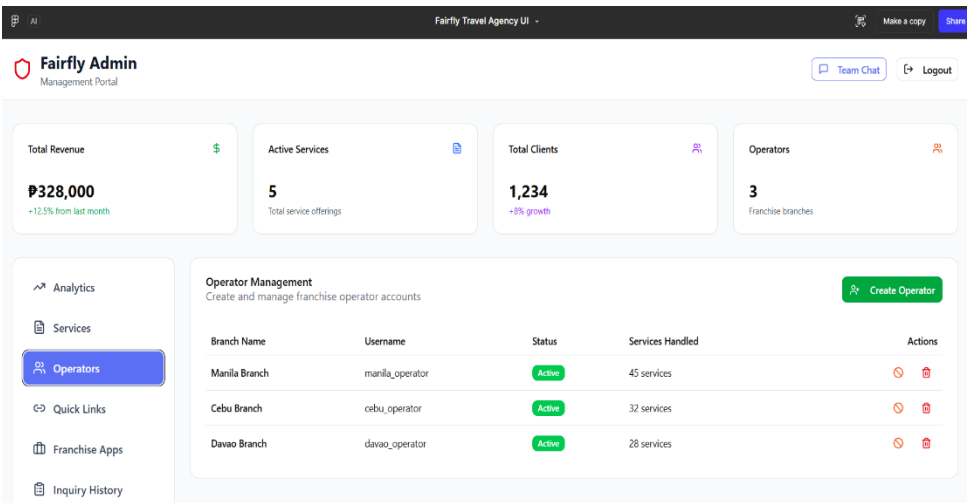
Service Management Page



The administrative interface for managing the collection of service templates available to franchise operators.

Figure 30

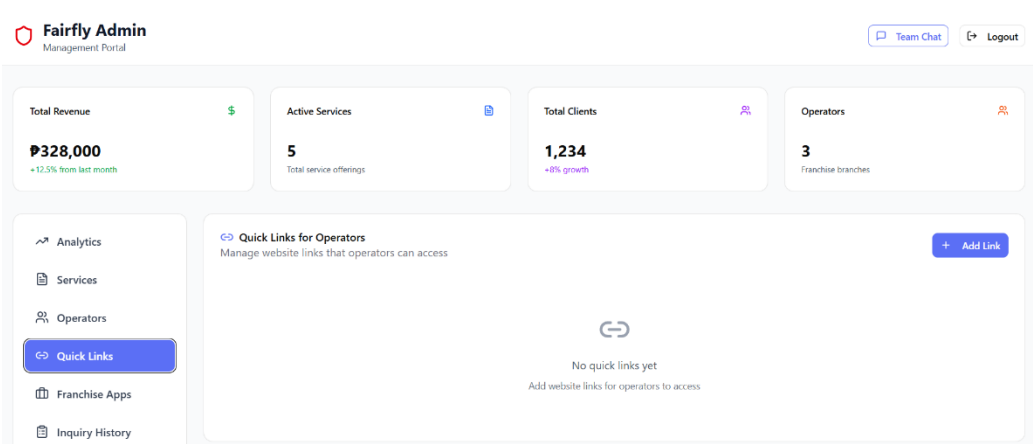
Franchise Management Page



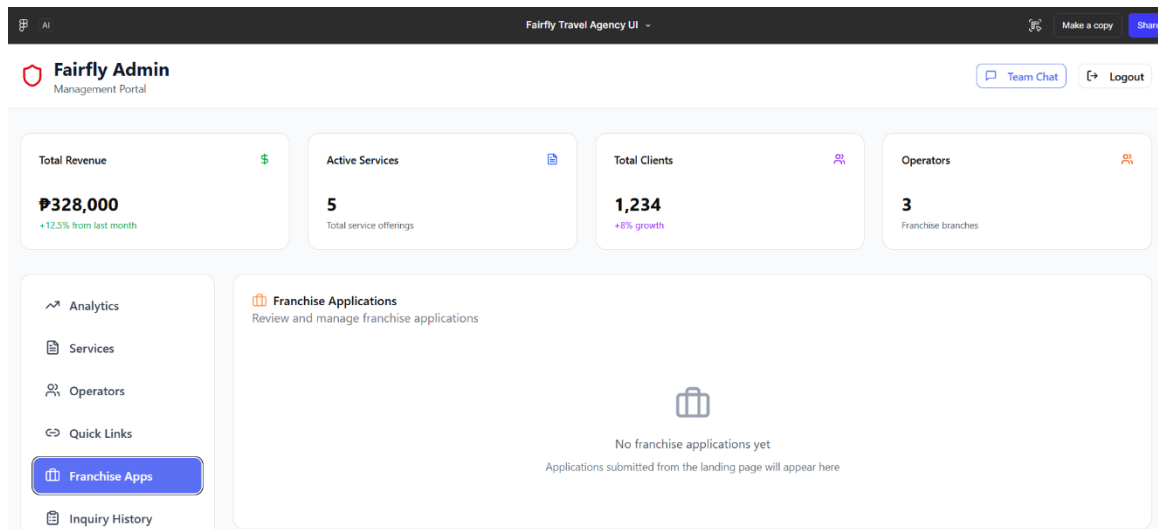
The administrative interface for registering, activating, and suspending franchise branch accounts.

Figure 31

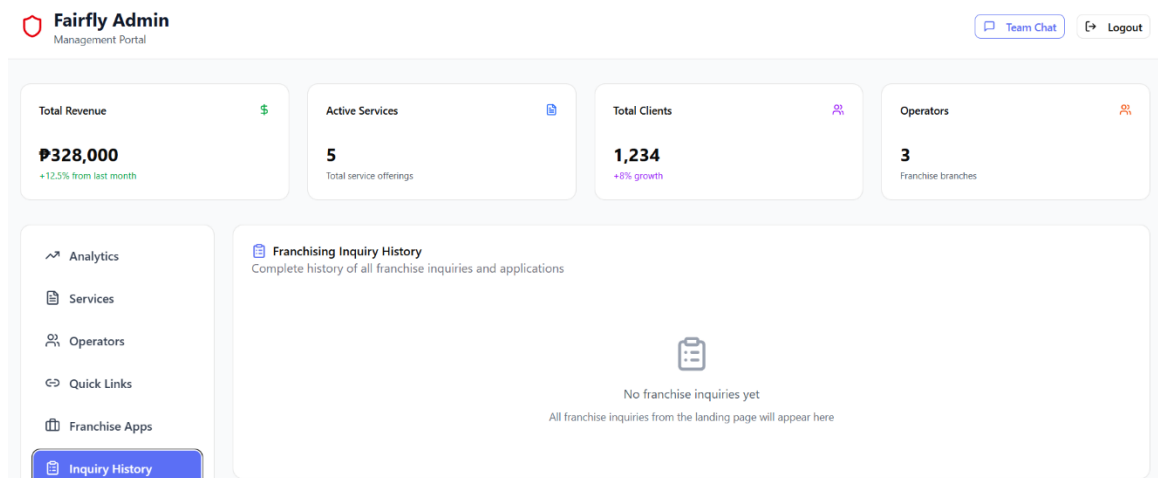
Admin Quick Link Page



The administrative interface for adding third-party website links that operators from other branch can use to complete services.

Figure 32*Admin Franchise Application Page*

The interface for displaying, confirming, and cancelation of application and appointment request from users that wants to franchise Fairfly.

Figure 33*Admin History Page*

The interface for displaying completed and cancelled application and appointment request from users that wants to franchise Fairfly.

Ethical Considerations

This study adheres to established ethical principles, particularly respect for persons, beneficence, and justice, ensuring that all participants are treated with dignity and their rights are fully protected throughout the research process. Prior to data collection, the research protocol underwent review and was granted ethical clearance by the College of Information and Communications Technology Local Ethics Review Committee (CICT-LERC) under the approval code BSIT-2026-0001. Participation in the study is strictly voluntary, and informed consent was obtained from all participants before their involvement. Participants were provided with sufficient information regarding the purpose of the study, procedures, duration, and their right to withdraw at any time without penalty or consequences.

To ensure confidentiality and data privacy, no personally identifiable information was collected unless necessary, and all data were anonymized or coded during analysis. Collected data were securely stored in password-protected digital storage accessible only to the researchers and their adviser, and will be retained only for 3 years before proper disposal. The study involved minimal risk, as it primarily focused on system evaluation and user feedback. No sensitive personal information was solicited, and participants were allowed to skip any questions they found uncomfortable. All procedures complied with institutional ethical standards and applicable data privacy regulations to safeguard participants' welfare and uphold the integrity of the research.

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